

SMART INDIA HACKATHON 2024 • COMPREHENSIVE ARCHITECTURE AUDIT

SamadhanSetu

Complete Current Project Analysis: Full Technical, Functional, Architecture, AI/ML, Database, API, Security & Implementation Report

TARGET ECOSYSTEM

Jharkhand Civic Crowdsourcing & Multi-Stakeholder Collaboration

AUDIT EXECUTION DATE

September 10, 2026

AUTOMATED TEST SUITES

16 / 16 Suites Passed (100% Verification, 695+ Assertions)

REPOSITORY WORKSPACE

d:ProjectsCrowdsource

OPERATIONAL SERVICES

All 5 Services Operational (Postgres, NestJS, Next.js, FastAPI, Expo)

DATABASE ENTITIES & MIGRATIONS

52 TypeORM Entities • 15 Sequential Migrations Applied

NestJS 10.3 API Gateway

FastAPI AI Microservice (Port 8000)

Next.js 14 App Router

Expo SDK 57 Mobile App

9 Regional Languages (OI Chiki, Devanagari)

PostgreSQL Advisory Lock Deduplication

7-Factor Trust-Weighted Hybrid Matching

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Complete structural inventory of technical findings, code references, database entities, and test results.

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Comprehensive Codebase Audit & Implementation State Report: SamadhanSetu

****Audit Execution Date:**** September 10, 2026 ****Source of Truth:**** Active source code, TypeORM entities, NestJS backend modules, FastAPI AI microservice, Next.js web portal, Expo SDK 57 mobile application, and automated test suites in [d:\Projects\Crowdsource](file:///d:/Projects/Crowdsource).

SECTION 1

PROJECT OVERVIEW

Core Identifiers

- **Project Name:** SamadhanSetu (Internal Application Identifier: `SamadhanPlatform`)
- **Repository Layout:** Monorepo containing 4 functional layers:
 1. `backend/` : NestJS 10.3 API & PostgreSQL ORM
 2. `frontend/` : Next.js 14 Web Portal & Governance Dashboard
 3. `mobile/` : Expo SDK 57 React Native Mobile App
 4. `ai-service/` : FastAPI Python Microservice (NVIDIA NIM / Deterministic Mock)
 5. `database/` : Initialization DDL & SQL schemas

A. Simple Explanation (Judge-Facing)

SamadhanSetu is an end-to-end crowdsourcing, problem intelligence, and multi-stakeholder collaborative solution platform built specifically for grassroots civic problem resolution in Jharkhand and regional India.

Citizens in rural villages and urban wards report community grievances—such as broken drinking water borewells, damaged bridges, power transformer failures, or agricultural infestations—using a mobile wizard supporting native languages and scripts (including Hindi, Santali in Ol Chiki script, and Nagpuri).

Instead of routing these as isolated complaints that get lost in administrative backlogs, the platform's AI pipeline instantly structures the problem, groups duplicate or localized reports into consolidated **Problem Clusters**, and calculates a transparent priority score. Local district government officers verify the problem cluster on their dashboard. Once verified, the problem opens for solution proposals: universities (like BIT Mesra or BirsA Agricultural University) and industry partners (like Tata Steel) submit Expressions of Interest (EOIs) detailing technical proposals, student/faculty teams, and CSR funding. The government forms an approved consortium project, monitors milestone execution through an immutable deliverable vault, and verifies real-world impact (population reached, health improvements) alongside innovation outcomes like filed patents and startups.

B. Technical Explanation (Architecture Implementation)

The platform is implemented as a microservice-orchestrated system:

1. **Intake & Draft Vault:** Citizens submit geotagged problem reports with multimedia evidence via Expo Mobile or Next.js Web to `ChallengesController`.
2. **AI Structuring & Multilingual Pipeline:** Upon submission, `AIAnalysisService` executes language detection and translation with a strict 3000ms bounded timeout. It queries the FastAPI `ai-service` to extract structured problem intelligence, normalizes required technical capabilities against an authoritative taxonomy in PostgreSQL, and generates 2048-dimensional vector embeddings stored in `entity_embeddings`.
3. **Deterministic Clustering & Advisory Locking:** `ProblemClustersService` acquires a PostgreSQL Transaction Advisory Lock (`pg_advisory_xact_lock`) scoped by district. It checks GPS distance (Haversine ≤ 8 km) and lexical Jaccard similarity. If similarity ≥ 0.75 , it auto-merges the report into an existing `ProblemCluster`; if between 0.50 and 0.74, it flags it as a `POTENTIAL_MATCH` for human triage; otherwise it instantiates an `INDEPENDENT` cluster.
4. **Human-in-the-Loop Verification Boundary:** AI recommendations and solution matching are strictly prevented while a problem is in draft or unverified status. Only when a Government Officer (`GOVERNMENT_OFFICER` or `GOVERNMENT_ADMIN`) reviews and validates the cluster (`POST /api/problem-clusters/:id/verify`) do linked reports become `VALIDATED` and trigger `MatchingService`.
5. **Hybrid Recommendation Engine:** `MatchingService` computes a weighted composite match across Semantic Similarity (25%), Capability Match with Trust Tiers (30%), Domain Expertise (20%), Geographic Proximity (10%), Verification Status (10%), Availability TTL Freshness (5%), and Novelty Exploration (+4%). Top recommendations are snapped into `recommendation_runs` and generate review records in `recommendation_reviews`.
6. **Consortium Formation & Project Governance:** Approved HEIs submit EOIs (`EoisController`). Administrators bundle multiple accepted EOIs into a single collaborative `Project`. Milestone progress triggers cascading locks on tasks and deliverables (`ProjectsService`).
7. **Impact Verification & Outcomes:** Completed projects submit structured impact metrics and field proof (`ImpactService`). Verified projects capture patent applications, startups, and technology transfers (`ProjectInnovationOutcome`).

SECTION 2

COMPLETE FEATURE INVENTORY

Core Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation
Citizen Challenge Creation	Multi-step wizard to report civic issues with geotagging and photo/document uploads	'CITIZEN'	['app/challenges/new/page.tsx' (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/new/page.tsx), ['mobile/app/report/new.tsx'] (file:///d:/Projects/Crowdsourc/mobile/app/report/new.tsx)	['ChallengesService'] (file:///d:/Projects/Crowdsourc/backend/src/modules/challenges/challenges.service.ts)
Challenge Submission & Intake	Transitions draft to active problem pipeline, triggering automated analysis	'CITIZEN'	['app/challenges/new/page.tsx' (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/new/page.tsx), ['mobile/app/report/new.tsx'] (file:///d:/Projects/Crowdsourc/mobile/app/report/new.tsx)	['ChallengesService.submitChallenge'] (file:///d:/Projects/Crowdsourc/backend/src/modules/challenges/challenges.service.ts#L112)
Public Challenge Discovery	Public directory of submitted, under-review, and validated civic problems	Anyone / Authenticated	['app/challenges/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/page.tsx), ['mobile/app/(tabs)/index.tsx'] (file:///d:/Projects/Crowdsourc/mobile/app/(tabs)/index.tsx)	['ChallengesService.getPublicChallenges'] (file:///d:/Projects/Crowdsourc/backend/src/modules/challenges/challenges.service.ts#L113)
Community Confirmation ("Me Too")	Citizens upvote/confirm recurring issues in their locality	'CITIZEN'	['app/challenges/[id]/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/[id]/page.tsx)	['ChallengesService.confirmChallenge'] (file:///d:/Projects/Crowdsourc/backend/src/modules/challenges/challenges.service.ts#L114)

AI Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation	API Endpoints
Problem Intelligence Extraction	Extracts root cause factors, domain, severity, and required technical skills	System / Automated	Displayed on ['app/reviewer-queue/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/reviewer-queue/page.tsx) & ['challenges/[id]/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/[id]/page.tsx)	['AiAnalysisService.analyzeChallenge'] (file:///d:/Projects/Crowdsourc/backend/src/modules/ai-analysis/ai-analysis.service.ts#L72), ['ai-service/main.py'] (file:///d:/Projects/Crowdsourc/ai-service/main.py#L77)	'POST /v1/ai/analyze-challenge', 'GET /api/ai/challenges/ic'
Taxonomy Normalization	Maps unstructured technical terms to canonical platform capabilities	System / Automated	Displayed in AI Analysis card	['AiAnalysisService.normalizeCapabilities'] (file:///d:/Projects/Crowdsourc/backend/src/modules/ai-analysis/ai-analysis.service.ts#L293), ['ai-service/app/taxonomy.py'] (file:///d:/Projects/Crowdsourc/ai-service/app/taxonomy.py)	'POST /v1/ai/normali-taxonomy'
Vector Embedding Generation	Produces semantic embeddings for similarity comparison	System / Automated	Displayed as vector provenance metadata	['AiAnalysisService.generateAndStoreChallengeEmbedding'] (file:///d:/Projects/Crowdsourc/backend/src/modules/ai-analysis/ai-analysis.service.ts#L337), ['ai-service/app/providers/nvidia.py'] (file:///d:/Projects/Crowdsourc/ai-service/app/providers/nvidia.py#L141)	'POST /v1/ai/embedd'
Concurrency-Safe Clustering	Deduplicates and consolidates similar reports into unified problem clusters	System / Automated	Filterable queue in ['ProblemClustersQueue.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/reviewer-queue/ProblemClustersQueue.tsx)	['ProblemClustersService.clusterCitizenReport'] (file:///d:/Projects/Crowdsourc/backend/src/modules/problem-clusters/problem-clusters.service.ts#L64)	'GET /api/problem-cl', 'GET /api/problem-clusters/id'
Multilingual Detection & Translation	Identifies regional languages/scripts and translates civic issues to English	'CITIZEN', Reviewers	Language Selector & On-demand translate button	['AiAnalysisService.detectLanguage'] (file:///d:/Projects/Crowdsourc/backend/src/modules/ai-analysis/ai-analysis.service.ts#L463), ['translateText'] (file:///d:/Projects/Crowdsourc/backend/src/modules/ai-analysis/ai-analysis.service.ts#L504)	'POST /v1/ai/detect-language', 'POST /v1/ai/translate', 'PO /api/challenges/id/tr'

User Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation	API Endpoints
User Authentication & JWT Session	Registration, login, profile fetch, password hashing	All Roles	['app/login/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/login/page.tsx), ['mobile/app/(auth)/login.tsx'] (file:///d:/Projects/Crowdsourc/mobile/app/(auth)/login.tsx)	['AuthController'] (file:///d:/Projects/Crowdsourc/backend/src/modules/auth/auth.controller.ts), ['AuthService'] (file:///d:/Projects/Crowdsourc/backend/src/modules/auth/auth.service.ts)	'POST /', 'POST /', 'GET /api', 'PATCH'
My Challenges Portal	Citizen view of draft, submitted, and validated personal reports	'CITIZEN'	['app/my-challenges/page.tsx'] (file:///d:/Projects/Crowdsourc/frontend/src/app/my-challenges/page.tsx), ['mobile/app/(tabs)/reports.tsx'] (file:///d:/Projects/Crowdsourc/mobile/app/(tabs)/reports.tsx)	['ChallengesService.getMyChallenges'] (file:///d:/Projects/Crowdsourc/backend/src/modules/challenges/challenges.service.ts#L112)	'GET /api'

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation	API Endpoints
Notification Center	In-app alerts for problem verification, review decisions, and milestone actions	All Roles	[mobile/app/(tabs)/notifications.tsx] (file:///d:/Projects/Crowdsourc/mobile/app/(tabs)/notifications.tsx)	[NotificationsController] (file:///d:/Projects/Crowdsourc/backend/src/modules/notifications/notifications.controller.ts)	'GET /api/notifications' 'PATCH /api/notifications/{id}' 'all-read /api/notifications'

Admin / Government Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation
Consolidated Cluster Verification	Government validation of problem clusters and all underlying citizen reports	'GOVERNMENT_OFFICER', 'GOVERNMENT_ADMIN', 'PLATFORM_ADMIN'	[ProblemClustersQueue.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/reviewer-queue/ProblemClustersQueue.tsx)	[ProblemClustersService.verifyCluster] (file:///d:/Projects/Crowdsourc/backend/src/modules/problem-clusters/problem-clusters.service.ts#L650)
Cluster Rejection with Audit Reason	Formal rejection of spurious problem clusters	'GOVERNMENT_OFFICER', 'GOVERNMENT_ADMIN', 'PLATFORM_ADMIN'	Rejection modal in [ProblemClustersQueue.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/reviewer-queue/ProblemClustersQueue.tsx)	[ProblemClustersService.rejectCluster] (file:///d:/Projects/Crowdsourc/backend/src/modules/problem-clusters/problem-clusters.service.ts#L711)
Potential Match Triage	Human evaluation of medium-confidence AI cluster suggestions	'GOVERNMENT_OFFICER', 'GOVERNMENT_ADMIN', 'PLATFORM_ADMIN'	Review card in [ProblemClustersQueue.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/reviewer-queue/ProblemClustersQueue.tsx)	[ProblemClustersService.reviewPotentialMatch] (file:///d:/Projects/Crowdsourc/backend/src/modules/problem-clusters/problem-clusters.service.ts#L553)
State Executive Analytics Dashboard	Macro and district-level KPIs, pipeline funnel, cluster status breakdown	'GOVERNMENT_ADMIN', 'PLATFORM_ADMIN', 'GOVERNMENT_OFFICER'	[app/government-dashboard/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/government-dashboard/page.tsx)	[AnalyticsController] (file:///d:/Projects/Crowdsourc/backend/src/modules/analytics/analytics.controller.ts)
EOI Review & Project Formation	Reviews institution EOIs and bundles accepted proposals into one collaborative project	'GOVERNMENT_OFFICER', 'GOVERNMENT_ADMIN', 'PLATFORM_ADMIN'	[app/challenges/[id]/eoi/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/[id]/eoi/page.tsx) & Admin EOI views	[AdminEoisController] (file:///d:/Projects/Crowdsourc/backend/src/modules/eois/admin-eois.controller.ts) [AdminChallengeProjectsController] (file:///d:/Projects/Crowdsourc/backend/src/modules/eois/admin-eois.controller.ts#L107)
Universal Verification Queue	Cross-organization audit queue for claims, capability evidence, and onboarding	'PLATFORM_ADMIN', 'GOVERNMENT_ADMIN', 'GOVERNMENT_OFFICER'	Admin Verification section	[AdminVerificationController] (file:///d:/Projects/Crowdsourc/backend/src/modules/verification/admin-verification.controller.ts)

Organization / Institution Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation
Capability Passport	Comprehensive institutional profile with departments, labs, faculty, and capabilities	'UNIVERSITY_ADMIN', 'INDUSTRY_ADMIN', 'PLATFORM_ADMIN'	[app/organizations/[id]/passport/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/organizations/[id]/passport/page.tsx)	[PassportController] (file:///d:/Projects/Crowdsourc/backend/src/modules/organizations/passport/passport.controller.ts)
Availability Freshness (TTL)	Capacity declaration and 14-day freshness renewal	Organization Admin	Passport availability toggle	[AvailabilityController] (file:///d:/Projects/Crowdsourc/backend/src/modules/organizations/passport/passport.controller.ts)
Expression of Interest (EOI) Submission	Universities/researchers apply to solve validated problems	'UNIVERSITY_ADMIN', 'FACULTY', 'INDUSTRY_ADMIN'	[app/challenges/[id]/eoi/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/challenges/[id]/eoi/page.tsx)	[ChallengeEoisController] (file:///d:/Projects/Crowdsourc/backend/src/modules/eois/challenge-eois.controller.ts) [EoisService] (file:///d:/Projects/Crowdsourc/backend/src/modules/eois/eois.service.ts)
Project Collaborative Workspace	Milestone tracking, task assignment, deliverable vault, blocker reporting	Consortium Members & Officers	[app/projects/[id]/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/projects/[id]/page.tsx)	[ProjectsController] (file:///d:/Projects/Crowdsourc/backend/src/modules/projects/projects.controller.ts)
Academic Collaboration Module	Multidisciplinary academic roster (students, faculty mentors, coordinators)	'UNIVERSITY_ADMIN', 'FACULTY', 'PLATFORM_ADMIN'	Project academic tab in [app/projects/[id]/page.tsx] (file:///d:/Projects/Crowdsourc/frontend/src/app/projects/[id]/page.tsx)	[ProjectsController] (file:///d:/Projects/Crowdsourc/backend/src/modules/projects/projects.controller.ts)

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation
Industry Contributions & CSR Co-Funding	Structured resource commitments (funding, hardware, prototyping, testing)	'INDUSTRY_ADMIN', 'INDUSTRY_MEMBER'	Project contributions tab	['ProjectsController'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/project)
Innovation & IP Outcomes Tracker	Records patents, startups, and technology transfers resulting from project	Consortium Lead & Admins	Project innovations tab	['ProjectsController'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/project)

Security & Governance Features

Feature Name	Purpose	User Role	Frontend Implementation	Backend Implementation
District Jurisdiction Isolation	Strictly restricts government officers to their designated district	'GOVERNMENT_OFFICER'	Automatic district filter badge	['JurisdictionService'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/services/jurisdiction.s)
Role-Based Access Control (RBAC)	Method-level route authorization guard	All Roles	Route guards in ['Navbar'] (file:///d:/Projects/Crowdsourcing/frontend/src/app/components/navbar.tsx)	['RolesGuard'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/guards/roles.guard.ts) ['@Roles()'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/decorators/roles.decorator.ts)
Deliverable & Evidence Storage Vault	MIME-type validated, isolated directory storage for uploads	All Roles	File input components	['EvidenceService'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/challenges/services/evidence)

SECTION 3

USER ROLES & RBAC

The platform defines 9 distinct user roles in `UserRole`.

TYPESCRIPT

```
export enum UserRole {
  CITIZEN = 'CITIZEN',
  UNIVERSITY_ADMIN = 'UNIVERSITY_ADMIN',
  FACULTY = 'FACULTY',
  STUDENT = 'STUDENT',
  INDUSTRY_ADMIN = 'INDUSTRY_ADMIN',
  INDUSTRY_MEMBER = 'INDUSTRY_MEMBER',
  GOVERNMENT_ADMIN = 'GOVERNMENT_ADMIN',
  GOVERNMENT_OFFICER = 'GOVERNMENT_OFFICER',
  PLATFORM_ADMIN = 'PLATFORM_ADMIN',
}
```

Detailed Role Definitions & Permissions

1. 'CITIZEN'

- Purpose:** Grassroots community member submitting societal problems and observing resolutions.
- Accessible Pages:** `/challenges`, `/challenges/new`, `/my-challenges`, `/challenges/:id`, Mobile app (all citizen screens).
- Allowed APIs:**
 - `POST /api/challenges` (create draft)
 - `PATCH /api/challenges/:id` (update draft)
 - `DELETE /api/challenges/:id` (delete draft)
 - `POST /api/challenges/:id/submit` (submit challenge)
 - `POST /api/challenges/:id/confirm` & `DELETE /api/challenges/:id/confirm`
 - `POST /api/challenges/:id/evidence`
 - `POST /api/projects/:id/impact/feedback` (submit citizen feedback if original author)
- Restricted Actions:** Cannot view reviewer queues, cannot verify clusters, cannot view administrative analytics, cannot form projects, cannot submit EOIs.

2. 'GOVERNMENT_OFFICER'

- Purpose:** District-level administrative reviewer responsible for verifying grassroots problem clusters and overseeing project execution within an assigned district.
- Jurisdiction:** Scoped strictly to `district_id` (`JurisdictionScope.DISTRICT`).
- Accessible Pages:** `/reviewer-queue`, `/government-dashboard`, `/challenges/:id`, `/projects/:id`.
- Allowed APIs:**
 - `GET /api/challenges/review/queue` (filtered by officer's district)
 - `GET /api/problem-clusters` (filtered by district)
 - `POST /api/problem-clusters/:id/verify` & `POST /api/problem-clusters/:id/reject`
 - `POST /api/problem-clusters/potential-matches/:id/review`
 - `GET /api/admin/analytics/*` (scoped to officer's district)
 - `GET /api/admin/eois`, `POST /api/admin/eois/:id/accept`
 - `POST /api/admin/challenges/:challengeId/projects`
 - `POST /api/admin/projects/:id/milestones/:milestoneId/review`
 - `POST /api/admin/projects/:id/complete`
- Restricted Actions:** Cannot verify challenges or clusters outside their district; cannot revoke verified impact (Platform Admin only); cannot modify system master capabilities.

3. 'GOVERNMENT_ADMIN'

- Purpose:** State-level government authority with oversight across all 24 districts of Jharkhand.
- Jurisdiction:** `JurisdictionScope.STATE`.
- Accessible Pages:** All reviewer and analytics pages across the state.
- Allowed Actions:** All actions of `GOVERNMENT_OFFICER` across all districts; can view statewide macro trends and assign district jurisdictions.

4. 'PLATFORM_ADMIN'

- Purpose:** Complete platform governance, system configuration, master taxonomy maintenance, and audit emergency actions.
- Jurisdiction:** `JurisdictionScope.NATIONAL`.
- Accessible Pages:** All pages across the entire platform.
- Unique Actions:**
 - `PATCH /api/auth/admin/users/:id/jurisdiction` (reassign officer jurisdictions)
 - `POST /api/admin/impact/:id/revoke` (audit revocation of verified impact)
 - `POST /api/admin/taxonomy-requests/:id/approve` & reject
 - `POST /api/admin/verification/:id/approve` & reject (claims, onboarding)

5. 'UNIVERSITY_ADMIN' & 'FACULTY'

- Purpose:** Higher education leadership and researchers proposing solutions to verified problems.
- Accessible Pages:** `/challenges`, `/challenges/:id`, `/challenges/:id/eoi`, `/my-eois`, `/organizations/:id/passport`, `/projects/:id`.
- Allowed Actions:**
 - Update organization capability passport
 - Submit, update, and withdraw EOIs on validated challenges
 - Manage academic members on active consortium projects
 - Submit milestone deliverables and progress updates

6. 'INDUSTRY_ADMIN' & 'INDUSTRY_MEMBER'

- Purpose:** Corporate, startup, and CSR partners offering financial co-funding, equipment, pilot sites, and manufacturing assistance.
- Accessible Pages:** Same as University, with focus on project contributions tab.
- Allowed Actions:** Propose and update project contributions (funding, prototyping, mentorship).

RBAC Permission Matrix

Operation / Endpoint	CITIZEN	GOVT_OFFICER	GOVT_ADMIN	PLATFORM_ADMIN	UNIVERSITY	INDUSTRY
Create & Submit Challenge	✓	✗	✗	✗	✗	✗
View Public Challenges	✓	✓	✓	✓	✓	✓
Confirm Challenge ("Me Too")	✓	✓	✓	✓	✓	✓
Access Reviewer Queue	✗	✓ (District)	✓ (State)	✓ (All)	✗	✗
Verify / Reject Problem Cluster	✗	✓ (District)	✓ (State)	✓ (All)	✗	✗
Submit EOI	✗	✗	✗	✗	✓	✓
Accept / Discuss EOI	✗	✓	✓	✓	✗	✗
Form Collaborative Project	✗	✓	✓	✓	✗	✗
Upload Milestone Deliverables	✗	✗	✗	✗	✓ (Lead)	✓ (Partner)
Review & Approve Milestones	✗	✓	✓	✓	✗	✗
Propose Industry Contribution	✗	✗	✗	✗	✗	✓
Submit Impact Assessment	✗	✗	✗	✗	✓ (Lead)	✗
Verify Impact Assessment	✗	✓	✓	✓	✗	✗
Revoke Impact Assessment	✗	✗	✗	✓ Only	✗	✗
Update Capability Passport	✗	✗	✗	✓	✓ (Own)	✓ (Own)
Manage Master Taxonomy	✗	✗	✗	✓ Only	✗	✗

SECTION 4

COMPLETE USER JOURNEYS

Journey 1: Citizen Problem Submission to Automated Clustering

```
Citizen (Mobile / Web)
├
├ [Form: Title, Description, Category, District, Block, Locality, GPS, Evidence]
├
├ POST /api/challenges (Status: DRAFT)
├
├ ChallengesController.createDraft()
├
├ ChallengesService.createDraft()
├
├ Database: Saved in "challenges" table (status = 'DRAFT')
├
├ Citizen reviews and clicks "Submit Problem"
├
├ POST /api/challenges/:id/submit
├
├ ChallengesController.submitChallenge()
├
├ ChallengesService.submitChallenge()
├
├ ┌ 1. Rate Limit & Field Validation Check (>=5 char title, >=10 char desc, district/block required)
├ ┌ 2. Status updated to 'SUBMITTED' in DB
├ ┌
├ ┌ 3. AiAnalysisService.analyzeChallenge() [Hard 3000ms bound]
├ ┌ ┌ Language Detection (Devanagari, Ol Chiki, Latin)
├ ┌ ┌ Translation to English if non-English (threshold >= 0.75)
├ ┌ ┌ AI Microservice call (/v1/ai/analyze-challenge)
├ ┌ ┌ ┌ Extracts: domain, summary, priority score, severity, required capabilities
├ ┌ ┌ ┌ Normalizes capabilities against master taxonomy
├ ┌ ┌ ┌ Saves ChallengeAiAnalysis record
├ ┌ ┌ ┌ Generates and stores 2048d vector embedding in "entity_embeddings"
├ ┌ ┌
├ ┌ ┌ 4. ProblemClustersService.clusterCitizenReport()
├ ┌ ┌ ┌ Acquires PostgreSQL Transaction Advisory Lock: pg_advisory_xact_lock(hashtext('cluster_lock:district'))
├ ┌ ┌ ┌ Filters candidate clusters in same district
├ ┌ ┌ ┌ Computes GPS Haversine distance (must be <= 8.0 km)
├ ┌ ┌ ┌ Computes Jaccard keyword similarity + locality bonus (0.25) + category bonus (0.10)
├ ┌ ┌ ┌
├ ┌ ┌ ┌ If Similarity >= 0.75: Auto-clusters with existing ProblemCluster; updates report count & priority
├ ┌ ┌ ┌ If 0.50 <= Similarity < 0.75: Marks clustering_status = 'POTENTIAL_MATCH' for human review
├ ┌ ┌ ┌ If Similarity < 0.50: Creates new independent ProblemCluster (status: AWAITING_GOVERNMENT_VERIFICATION)
├ ┌ ┌
├ ┌ ┌ 5. Dispatches in-app notification to District Officers
├ ┌
├ HTTP 200 Response with verification_display_status = 'Pending Government Verification'
├
├ Frontend displays confirmation screen with Cluster ID and Tracking Number
```

Journey 2: Government Verification & Recommendation Generation

```
Government Officer (Web Portal)
├
├ Navigates to /reviewer-queue
├
├ GET /api/problem-clusters?status=AWAITING_GOVERNMENT_VERIFICATION
├
├ ProblemClustersController -> ProblemClustersService.getClusters()
├
├ Officer inspects Problem Cluster:
├ ┌ - Consolidated citizen reports list
├ ┌ - Aggregated photographic/video evidence
├ ┌ - AI extracted root causes & required technical capabilities
├ ┌ - Priority Score breakdown (severity, count, evidence, geographic concentration)
├ ┌
├ Officer clicks "Verify & Open for Solutions"
├
├ POST /api/problem-clusters/:id/verify
├
├ ProblemClustersController.verifyCluster()
├
├ ProblemClustersService.verifyCluster()
├ ┌ Updates ProblemCluster status to 'VALIDATED'
├ ┌ Updates all linked Challenges status to 'VALIDATED' (atomically)
├ ┌
├ ┌ Triggers Post-Verification Matching Engine:
├ ┌ ┌ MatchingService.generateRecommendations(challengeId)
├ ┌ ┌
├ ┌ ┌ ┌ Retrieves all verified & unverified Organizations (skips REJECTED)
├ ┌ ┌ ┌ Computes Hybrid Score:
├ ┌ ┌ ┌ ┌ Semantic Vector Similarity (25%)
├ ┌ ┌ ┌ ┌ Trust-Weighted Capability Match (30%):
├ ┌ ┌ ┌ ┌ ┌ [VERIFIED = 1.0, PENDING = 0.65, UNVERIFIED = 0.35, REJECTED = 0.0]
├ ┌ ┌ ┌ ┌ ┌ Domain Expertise (20%) [HEI Labs/Faculty vs Industry Support Types]
├ ┌ ┌ ┌ ┌ ┌ Geographic Relevance (10%) [National vs Statewide vs District]
├ ┌ ┌ ┌ ┌ ┌ Verification Confidence (10%)
├ ┌ ┌ ┌ ┌ ┌ Availability TTL Freshness (5%) [Fresh = 1.0, Stale = 0.75]
├ ┌ ┌ ┌ ┌ ┌ Novelty Exploration Boost (+4% for new verified institutions)
├ ┌ ┌ ┌
├ ┌ ┌ ┌ Reranks Top 10 via /v1/ai/rerank
├ ┌ ┌ ┌ Saves immutable snapshot in "recommendation_runs"
├ ┌ ┌ ┌ Saves review records in "recommendation_reviews"
├ ┌ ┌
├ ┌ ┌ Dispatches targeted in-app notifications to matched Higher Education Institutions
├ ┌
├ HTTP 200 OK
├
├ Problem Cluster status updates to VALIDATED; matched universities receive notifications
```

Journey 3: Expression of Interest (EOI) to Collaborative Project Formation

```
University Dean / Faculty Researcher
├
└ Receives Notification / Browses /challenges -> Finds Validated Problem
  └
  └ Clicks "Submit Expression of Interest" -> Navigates to /challenges/:id/eoi
    └
    └ Submits proposal: technical approach, timeline, faculty mentor, student team, requested funding
      └
      └ POST /api/challenges/:id/eois (Creates EOI in DRAFT)
        └ POST /api/eois/:id/submit (Transitions to UNDER_REVIEW)
          └
          └ Government Officer opens /admin/eois
            └
            └ Reviews EOI proposal alongside AI match breakdown
              └
              └ Options:
                1. Request Discussion: POST /api/admin/eois/:id/request-discussion (Status -> DISCUSSION_REQUIRED)
                2. Reject: POST /api/admin/eois/:id/reject (Status -> REJECTED)
                3. Accept into Candidate Pool: POST /api/admin/eois/:id/accept (Status -> ACCEPTED)
                  └
                  └ [CRITICAL RULE: Accepting EOI NEVER creates a project]
                    └
                    └ Officer navigates to /admin/challenges/:id/projects
                      └
                      └ GET /api/admin/challenges/:id/accepted-eois
                        (Lists all accepted academic & industry EOIs)
                          └
                          └ Officer selects:
                            - Lead Academic Institution (e.g. BIT Mesra)
                            - Consortium Partner (e.g. Birsā Agricultural University)
                            - Industry/CSR Co-funder (e.g. Tata Steel)
                              └
                              └ POST /api/admin/challenges/:id/projects (FormCollaborativeProjectDto)
                                └
                                └ AdminChallengeProjectsController.formProject()
                                  └
                                  └ EoisService.formCollaborativeProject()
                                    └
                                    └ ┌ Creates Project entity (status: INITIATED / KICKOFF_PENDING)
                                      └
                                      └ ┌ Creates ProjectParticipant records with explicit roles (LEAD, PARTNER, FUNDER)
                                        └
                                        └ ┌ Links selected EOIs (status: PROJECT_FORMED)
                                          └
                                          └ ┌ Updates parent Challenge and ProblemCluster status to PROJECT_INITIATED (intake closed)
                                            └
                                            └ Consortium Project Workspace is live at /projects/:projectId
```

Journey 4: Project Execution, Milestone Governance & Impact Verification

```
Consortium Lead (BIT Mesra)
├
└ Submits Project Kickoff Plan: POST /api/projects/:id/kickoff
  └
  └ Officer Approves Kickoff: POST /api/admin/projects/:id/kickoff-review -> Project status: ACTIVE
    └
    └ Lead creates Milestones and Tasks: POST /api/projects/:id/milestones
      └ Uploads field photos, specs, lab reports: POST /api/projects/:id/deliverables
        └
        └ Lead requests Milestone Review:
          POST /api/projects/:id/milestones/:milestoneId/request-review
            └
            └ [CASCADING LOCK ENGAGES: Milestone, tasks, and deliverables become strictly READ-ONLY]
              └
              └ Officer evaluates deliverables:
                - If Revision: POST /api/admin/projects/:id/milestones/:id/review (REVISION_REQUIRED -> Unlocks)
                - If Approved: POST /api/admin/projects/:id/milestones/:id/review (APPROVED -> Permanently locked)
                  └
                  └ If all milestones approved, Officer completes project:
                    POST /api/admin/projects/:id/complete (Zero-Milestone Completion Guard verified)
                    Project status -> COMPLETED
                      └
                      └ Lead submits Impact Assessment:
                        POST /api/projects/:id/impact
                        Adds structured metrics (Population Reached, Cost Savings): POST /api/projects/:id/impact/metrics
                        Uploads verification evidence: POST /api/projects/:id/impact/evidence
                        Submits for verification: POST /api/projects/:id/impact/submit
                          └
                          └ Original Citizen Reporter submits feedback: POST /api/projects/:id/impact/feedback
                            └
                            └ Officer approves Impact: POST /api/admin/impact/:id/approve
                              Project status -> IMPACT_VERIFIED
                                └
                                └ Consortium logs Innovation Outcomes:
                                  POST /api/projects/:id/innovation-outcomes (Patents filed, Startups created, Tech transferred)
                                  Officer verifies outcomes: POST /api/admin/projects/:id/innovation-outcomes/:id/verify
```

SECTION 5 FRONTEND ANALYSIS

Framework & Build Configuration

- **Framework:** Next.js 14.2.35 (React 18.2.0)
- **Routing:** Next.js App Router (`frontend/src/app`)
- **Language:** TypeScript 5.3.3
- **Styling & CSS:** Tailwind CSS 3.4.1 + `tailwind-merge` + `clsx`
- **Icons:** `lucide-react` 0.312.0
- **State Management:** React Context API (`AuthContext` , `I18nContext`) + Component local state
- **API Client:** Native `fetch` with Bearer token injection from `localStorage`
- **i18n Implementation:** Custom React Context (`src/lib/i18n.tsx`) supporting 9 languages loaded from `src/locales/*.json` .

Folder & Module Structure

```
frontend/src/
├── app/
│   ├── layout.tsx           # Root layout with AuthProvider, I18nProvider, and Navbar
│   ├── page.tsx            # Landing page with platform overview & role portals
│   └── globals.css          # Tailwind base styles and CSS variables
├── (auth)/
│   └── login/page.tsx       # Universal role login page with preset demo accounts
├── challenges/
│   ├── page.tsx            # Public browse challenges with filters & search
│   ├── new/page.tsx        # Citizen challenge submission form with evidence upload
│   └── [id]/
│       ├── page.tsx        # Challenge detail, confirmations, and AI analysis card
│       └── eoi/page.tsx     # University / Industry EOI submission wizard
├── my-challenges/page.tsx   # Citizen dashboard of personal submitted issues
├── my-eois/page.tsx         # Institution dashboard of submitted EOIs
├── reviewer-queue/
│   ├── page.tsx            # Main reviewer workspace
│   └── ProblemClustersQueue.tsx # Consolidated Problem Cluster triage and verification
├── government-dashboard/page.tsx # State Executive Analytics Dashboard (KPIs, funnel, clusters)
├── organizations/
│   ├── onboard/page.tsx    # Organization onboarding proposal form
│   └── [id]/passport/page.tsx # Capability Passport view & editing
├── projects/[id]/page.tsx   # Consortium project workspace (milestones, tasks, deliverables)
├── impact/[projectId]/page.tsx # Impact assessment and community feedback page
├── components/
│   ├── navbar.tsx          # Responsive navigation with role-based links & language switcher
│   └── auth-card.tsx        # Preset credential cards for instant demo access
├── lib/
│   ├── auth-context.tsx    # JWT token storage, user state, and login/logout handlers
│   ├── i18n.tsx           # Multilingual translation hook t() and language switcher
│   └── utils.ts            # Date formatting (formatDateSafe) and role formatting helpers
└── locales/
    └── # 9 JSON dictionary files:
        ├── en.json, hi.json, sat.json, nag.json, mun.json, kru.json, kho.json, sad.json, pan.json
```

SECTION 6

BACKEND ANALYSIS

Architecture Pattern

- **Framework:** NestJS 10.3.0 on Node.js 20+
- **Language:** TypeScript 5.3.3
- **Design Pattern:** Modular Layered Architecture (Controller → Guard → Service → TypeORM Repository → PostgreSQL).
- **Cross-Cutting Concerns:**
 - Global Versioned Prefix: `/api` (`main.ts`)
 - Global Validation: `ValidationPipe` with `whitelist: true` , `transform: true` (`main.ts`)
 - Cross-Origin Resource Sharing: `app.enableCors({ origin: '*', credentials: true })`
 - Background Timers: `ScheduleModule` with `DraftCleanupService` running hourly to clean abandoned drafts (> 72 hours).

Request Lifecycle

```
HTTP Request (e.g. POST /api/challenges/123/submit)
|
Express Middleware / CORS / Body Parser
|
JwtAuthGuard (Extracts Bearer token, validates signature via JwtService, attaches req.user)
|
RolesGuard (Reflector reads @Roles() metadata, checks req.user.role in allowed roles)
|
ValidationPipe (Validates DTO using class-validator decorators, strips non-whitelisted properties)
|
Controller Handler (e.g. ChallengesController.submitChallenge())
|
Service Business Logic (e.g. ChallengesService.submitChallenge())
|
├── External Microservice Call: fetch('http://127.0.0.1:8000/v1/ai/analyze-challenge')
│   └── [Bounded by 3000ms hard timeout via AbortSignal.timeout(3000)]
|
├── Transaction Runner (e.g. queryRunner.startTransaction())
│   └── Advisory Lock: SELECT pg_advisory_xact_lock(hashtext('cluster_lock:ranchi'))
|
└── TypeORM Repository: challengeRepo.save(challenge)
|
Database: PostgreSQL Execution
|
DTO Serialization / Anonymization
|
HTTP Response (e.g. 200 OK JSON)
```

SECTION 7 API INVENTORY

Every single endpoint listed below is verified from the active backend controller source code:

1. Authentication & User APIs ([AuthController](file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/auth.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
POST	/api/auth/register	Register citizen or user account	No	Public	[RegisterDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/dto/register.dto.ts) (name, email, password, phone, language)	Safe User Object (no password)
POST	/api/auth/login	Authenticate user credentials	No	Public	[LoginDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/auth/dto/login.dto.ts) (email, password)	{ accessToken, user }
GET	/api/auth/me	Retrieve profile of authenticated user	Yes	Any Authenticated	None	User profile object
PATCH	/api/auth/me	Update personal profile (phone, preferred language)	Yes	Any Authenticated	Partial user fields	Updated User profile
PATCH	/api/auth/admin/users/:id/jurisdiction	Reassign officer district jurisdiction	Yes	PLATFORM_ADMIN	{ district_id, district, state, jurisdiction_scope }	Updated User profile
GET	/api/auth/admin-test	RBAC test verification endpoint	Yes	PLATFORM_ADMIN	None	{ message, user, timestamp }

2. Challenge & Crowdsourcing APIs ([ChallengesController](file:///d:/Projects/Crowdsourcing/backend/src/modules/challenges/challenges.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
POST	/api/challenges	Create draft challenge	Yes	Any Authenticated	[CreateChallengeDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/challenges/dto/create-challenge.dto.ts)	Created 'Challenge' (status: DRAFT)
GET	/api/challenges	Public challenge discovery (no drafts)	No	Public / Optional Auth	Query params (category, district, block, page)	{ items, total, page, limit }
GET	/api/challenges/my	View personal challenges (with drafts)	Yes	Any Authenticated	None	Array of 'Challenge'
GET	/api/challenges/review/queue	Reviewer queue of submitted challenges	Yes	GOVERNMENT_OFFICER, GOVT_ADMIN, PLATFORM_ADMIN	Query 'status'	Array of 'Challenge'
GET	/api/challenges/evidence/file/:filename	Stream uploaded evidence file safely	No	Public	URL param 'filename'	Binary file stream
GET	/api/challenges/:id	View challenge detail	No	Public / Optional Auth	URL param 'id'	'Challenge' with evidence and counts
POST	/api/challenges/:id/translate	On-demand translation to Hindi or target language	No	Public / Optional Auth	{ target_language }	{ id, target_language, translated_title, translated_description, cached }
PATCH	/api/challenges/:id	Update draft challenge	Yes	Author / PLATFORM_ADMIN	[UpdateChallengeDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/challenges/dto/update-challenge.dto.ts)	Updated 'Challenge'
DELETE	/api/challenges/:id	Delete draft challenge & files	Yes	Author / PLATFORM_ADMIN	None	{ success, message }
POST	/api/challenges/:id/submit	Submit draft challenge to intake pipeline	Yes	Author	None	Submitted 'Challenge' with AI & cluster
POST	/api/challenges/:id/confirm	Community confirmation ("Me Too")	Yes	Any Authenticated	None	{ confirmed: true, confirmationsCount }
DELETE	/api/challenges/:id/confirm	Remove community confirmation	Yes	Any Authenticated	None	{ confirmed: false, confirmationsCount }
POST	/api/challenges/:id/review	Administrative approval or rejection	Yes	GOVERNMENT_OFFICER, GOVT_ADMIN, PLATFORM_ADMIN	[ReviewChallengeDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/challenges/dto/review-challenge.dto.ts)	Reviewed 'Challenge'
POST	/api/challenges/:id/evidence	Upload evidence file (multipart)	Yes	Author	Multipart 'file', 'title', 'description'	Created 'ChallengeEvidence'
DELETE	/api/challenges/:id/evidence/:evidenceId	Remove evidence file from draft	Yes	Author	URL params	{ success: true }

3. Problem Clusters APIs ([`ProblemClustersController`](file:///d:/Projects/Crowdsource/backend/src/modules/problem-clusters/problem-clusters.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
GET	/api/problem-clusters	List problem clusters with filters	No	Public	Query params ('district', 'status', 'priority')	{ items, total, page, limit }
GET	/api/problem-clusters/id	Detailed cluster view with all child reports	No	Public	URL param 'id'	'ProblemCluster' + aggregated evidence + reports
POST	/api/problem-clusters/id/verify	Government verification of cluster	Yes	'GOVERNMENT_OFFICER', 'GOVT_ADMIN', 'PLATFORM_ADMIN'	None	Verified 'ProblemCluster'
POST	/api/problem-clusters/id/reject	Rejection of cluster with mandatory reason	Yes	'GOVERNMENT_OFFICER', 'GOVT_ADMIN', 'PLATFORM_ADMIN'	{ reason }	Rejected 'ProblemCluster'
POST	/api/problem-clusters/potential-matches/reportId/review	Triage AI medium-confidence potential match	Yes	'GOVERNMENT_OFFICER', 'GOVT_ADMIN', 'PLATFORM_ADMIN'	{ action: 'ACCEPT' 'REJECT' }	Result object with new or merged cluster

4. Expression of Interest (EOI) & Formation APIs ([`ChallengeEoisController`](file:///d:/Projects/Crowdsource/backend/src/modules/eois/eois.controller.ts), [`EoisController`](file:///d:/Projects/Crowdsource/backend/src/modules/eois/eois.controller.ts), [`AdminEoisController`](file:///d:/Projects/Crowdsource/backend/src/modules/eois/admin-eois.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
POST	/api/challenges/challengeld/eois	Create EOI in DRAFT	Yes	Org Representative	['CreateEoiDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/eois/dto/index.ts)	Created 'ExpressionOfInterest'
GET	/api/eois/my	View organization's EOIs	Yes	Org Representative	None	Array of 'ExpressionOfInterest'
GET	/api/eois/id	View EOI detail	Yes	Org Rep / Reviewer	URL param 'id'	'ExpressionOfInterest' detail
PATCH	/api/eois/id	Update EOI in DRAFT or DISCUSSION	Yes	Proposing Org	['UpdateEoiDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/eois/dto/index.ts)	Updated 'ExpressionOfInterest'
POST	/api/eois/id/submit	Submit DRAFT EOI for review	Yes	Proposing Org	None	Submitted 'ExpressionOfInterest'
POST	/api/eois/id/resubmit	Resubmit after clarification	Yes	Proposing Org	None	Resubmitted 'ExpressionOfInterest'
POST	/api/eois/id/withdraw	Withdraw EOI prior to formation	Yes	Proposing Org	{ reason }	Withdrawn 'ExpressionOfInterest'
POST	/api/eois/id/evidence	Upload supporting proposal document	Yes	Proposing Org	Multipart 'file'	Created 'EoiEvidence'
GET	/api/admin/eois	Reviewer queue of EOIs	Yes	Reviewer Roles	Query 'status', 'challengeld'	Array of 'ExpressionOfInterest'
POST	/api/admin/eois/id/request-discussion	Request clarification on proposal	Yes	Reviewer Roles	{ review_notes }	Updated 'ExpressionOfInterest'
POST	/api/admin/eois/id/accept	Accept EOI into candidate pool	Yes	Reviewer Roles	None	Accepted 'ExpressionOfInterest'
POST	/api/admin/eois/id/reject	Reject EOI with mandatory reason	Yes	Reviewer Roles	{ reason }	Rejected 'ExpressionOfInterest'
GET	/api/admin/challenges/challengeld/accepted-eois	List accepted EOIs for challenge	Yes	Reviewer Roles	URL param 'challengeld'	Array of accepted EOIs
POST	/api/admin/challenges/challengeld/projects	Form collaborative project from accepted EOIs	Yes	Reviewer Roles	['FormCollaborativeProjectDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/eois/dto/index.ts)	Created 'Project'

5. Project Execution & Governance APIs ([`ProjectsController`](file:///d:/Projects/Crowdsource/backend/src/modules/projects/projects.controller.ts), [`AdminProjectsController`](file:///d:/Projects/Crowdsource/backend/src/modules/projects/admin-projects.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
GET	/api/projects	List projects accessible to user	Yes	Any Authenticated	Query 'status'	Array of 'Project'
GET	/api/projects/id	View comprehensive project workspace	Yes	Consortium / Reviewer	URL param 'id'	Full 'Project' work
POST	/api/projects/id/kickoff	Submit kickoff plan	Yes	Consortium Lead	['ProjectKickoffDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/projects/dto/index.ts)	Updated 'Project'
POST	/api/admin/projects/id/kickoff-review	Approve or request revision on kickoff	Yes	Reviewer Roles	['KickoffReviewDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/projects/dto/index.ts)	Updated 'Project'
GET	/api/projects/id/milestones	List project milestones & deliverables	Yes	Consortium / Reviewer	URL param 'id'	Array of 'ProjectMilestone'
POST	/api/projects/id/milestones	Create milestone	Yes	Consortium Lead	['CreateMilestoneDto'] (file:///d:/Projects/Crowdsource/backend/src/modules/projects/dto/index.ts)	Created 'ProjectMilestone'

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
'PATCH'	'/api/projects/id/milestones/milestoneld'	Update milestone	Yes	Consortium Lead	['UpdateMilestoneDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Updated 'Project'
'POST'	'/api/projects/id/milestones/milestoneld/start'	Start pending milestone	Yes	Consortium Lead	None	Milestone in 'IN_PROGRESS'
'POST'	'/api/projects/id/milestones/milestoneld/request-review'	Submit milestone (Engages Lock)	Yes	Consortium Lead	None	Milestone in REVIEW_REQUEST
'POST'	'/api/admin/projects/id/milestones/milestoneld/review'	Approve or request revision	Yes	Reviewer Roles	['MilestoneReviewDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Milestone review
'GET'	'/api/projects/id/tasks'	List tasks	Yes	Consortium / Reviewer	Query 'milestoneld'	Array of 'ProjectTasks'
'POST'	'/api/projects/id/tasks'	Create task	Yes	Consortium Member	['CreateTaskDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Created 'ProjectTask'
'PATCH'	'/api/projects/id/tasks/taskld'	Update task status	Yes	Assignee / Lead	['UpdateTaskDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Updated 'ProjectTask'
'POST'	'/api/projects/id/deliverables'	Upload deliverable evidence (vault)	Yes	Consortium Member	Multipart 'file', 'document_type', etc.	Created 'ProjectDeliverable'
'GET'	'/api/projects/id/deliverables/deliverableld/download'	Download deliverable file	Yes	Consortium / Reviewer	URL params	Binary stream
'POST'	'/api/projects/id/updates'	Post progress update or blocker	Yes	Consortium Member	['CreateProjectUpdateDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Created 'ProjectUpdate'
'POST'	'/api/admin/projects/id/blockers/updateld/review'	Review or resolve blocker	Yes	Reviewer Roles	['BlockerReviewDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Blocker resolution
'POST'	'/api/admin/projects/id/complete'	Complete project (Zero-milestone guard)	Yes	Reviewer Roles	['ProjectCompletionDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Completed 'Project'
'POST'	'/api/admin/projects/id/terminate'	Terminate project with reason	Yes	Reviewer Roles	['ProjectTerminationDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/index.ts)	Terminated 'Project'
'GET'	'/api/projects/id/academic-members'	List academic team members	Yes	Consortium / Reviewer	URL param 'id'	Array of 'ProjectAcademicMembers'
'POST'	'/api/projects/id/academic-members'	Add student/faculty member	Yes	University Admin/Lead	['CreateAcademicMemberDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/academic-member.dto.ts)	Created member
'GET'	'/api/projects/id/contributions'	List industry contributions	Yes	Consortium / Reviewer	URL param 'id'	Array of 'ProjectContributions'
'POST'	'/api/projects/id/contributions'	Record industry contribution	Yes	Industry Partner	['CreateProjectContributionDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/project-contribution.dto.ts)	Created contribution
'POST'	'/api/admin/projects/contributions/contribld/verify'	Government verify contribution	Yes	Reviewer Roles	['VerifyContributionDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/project-contribution.dto.ts)	Verified contribution
'GET'	'/api/projects/id/innovation-outcomes'	List patents/startups generated	Yes	Consortium / Reviewer	URL param 'id'	Array of 'ProjectInnovationOutcomes'
'POST'	'/api/projects/id/innovation-outcomes'	Record patent or startup outcome	Yes	Consortium Lead	['CreateInnovationOutcomeDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/innovation-outcome.dto.ts)	Created outcome
'POST'	'/api/admin/projects/id/innovation-outcomes/outcomeld/verify'	Government verify IP outcome	Yes	Reviewer Roles	['VerifyInnovationOutcomeDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/projects/dto/innovation-outcome.dto.ts)	Verified outcome

6. Impact Verification APIs (['ImpactController'](file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/impact.controller.ts), ['AdminImpactController'](file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/admin-impact.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
'GET'	'/api/projects/id/impact'	View impact assessment & metrics	Yes	Any Authenticated	URL param 'id'	'ImpactAssessment' + metrics + evidence + feedback
'POST'	'/api/projects/id/impact'	Initiate impact assessment	Yes	Consortium Lead / Admin	['CreateImpactAssessmentDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Created 'ImpactAssessment'
'POST'	'/api/projects/id/impact/submit'	Submit impact for verification	Yes	Consortium Lead	None	Submitted 'ImpactAssessment'
'POST'	'/api/projects/id/impact/metrics'	Add impact metric (e.g. pop reached)	Yes	Consortium Lead	['CreateImpactMetricDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Created 'ImpactMetric'
'POST'	'/api/projects/id/impact/evidence'	Upload impact evidence file	Yes	Consortium Lead	Multipart 'file'	Created 'ImpactEvidence'
'POST'	'/api/projects/id/impact/feedback'	Citizen beneficiary feedback	Yes	Original Submitter	['CreateImpactFeedbackDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Created 'ImpactFeedback'
'GET'	'/api/admin/impact/queue'	Reviewer queue of impact submissions	Yes	Reviewer Roles	Query 'status'	Array of 'ImpactAssessment'
'POST'	'/api/admin/impact/id/approve'	Approve impact (transitions project to IMPACT_VERIFIED)	Yes	Reviewer Roles	['ImpactReviewActionDto'] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Approved assessment

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
POST	/api/admin/impact/:id/require-revision	Request impact revision	Yes	Reviewer Roles	[RequireRevisionDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Assessment in 'REVISION_REQUIRED'
POST	/api/admin/impact/:id/reject	Reject impact assessment	Yes	Reviewer Roles	[RejectImpactDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Rejected assessment
POST	/api/admin/impact/:id/revoke	Audit revocation (Platform admin only)	Yes	PLATFORM_ADMIN	[RevokeImpactDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/impact/dto/index.ts)	Reverted assessment

7. Capability Passport & Organization APIs ([PassportController](file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/controllers/passport.controller.ts), [AvailabilityController](file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/controllers/availability.controller.ts), [OnboardingController](file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/controllers/onboarding.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
GET	/api/organizations/:id/passport	View full capability passport	No	Public	URL param 'id'	Organization + sub-profiles + capabilities
PUT	/api/organizations/:id/passport	Update organization profile & labs	Yes	Org Representative	[UpdatePassportDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/dto/update-passport.dto.ts)	Updated passport
POST	/api/organizations/:id/capabilities	Attach capability claim to passport	Yes	Org Representative	[AddCapabilityDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/dto/add-capability.dto.ts)	Created capability claim
DELETE	/api/organizations/:id/capabilities/:capId	Detach capability from passport	Yes	Org Representative	URL params	{ success: true }
POST	/api/organizations/:id/evidence	Upload capability evidence doc	Yes	Org Representative	Multipart 'file' or 'external_url'	Created 'OrganizationEvidence'
GET	/api/organizations/:id/availability	Check capacity & TTL freshness	No	Public	URL param 'id'	{ available_capacity, availability_status, expires_at }
PUT	/api/organizations/:id/availability	Renew availability TTL (14 days)	Yes	Org Representative	{ available_capacity, ttl_days }	Confirmed availability
POST	/api/organizations/onboarding-requests	Propose new organization onboarding	Yes	Any Authenticated	[CreateOnboardingRequestDto] (file:///d:/Projects/Crowdsourcing/backend/src/modules/organizations/dto/create-onboarding-request.dto.ts)	Created request
POST	/api/admin/onboarding-requests/:id/approve	Approve onboarding request	Yes	PLATFORM_ADMIN, GOVT_OFFICER	{ admin_notes }	Created 'Organization'

8. Analytics & Supporting APIs ([AnalyticsController](file:///d:/Projects/Crowdsourcing/backend/src/modules/analytics/analytics.controller.ts), [NotificationsController](file:///d:/Projects/Crowdsourcing/backend/src/modules/notifications/notifications.controller.ts), [LocationsController](file:///d:/Projects/Crowdsourcing/backend/src/modules/locations/locations.controller.ts))

Method	Endpoint	Purpose	Auth Required	Allowed Roles	Request Body	Response
GET	/api/admin/analytics/overview	Executive KPI cards & funnel	Yes	Reviewer Roles	Query filters ('district', 'domain')	Overview statistics
GET	/api/admin/analytics/districts	District comparison metrics	Yes	Reviewer Roles	Query filters	District rankings & metrics
GET	/api/admin/analytics/action-queue	Urgency action queue for officers	Yes	Reviewer Roles	Query filters	High-priority pending items
GET	/api/admin/analytics/matching-insights	Capability matching distribution	Yes	Reviewer Roles	Query filters	Matching analytics
GET	/api/admin/analytics/problem-clusters	Cluster breakdown by domain	Yes	Reviewer Roles	Query filters	Clustering metrics
GET	/api/admin/analytics/projects	Collaborative project governance metrics	Yes	Reviewer Roles	Query filters	Project stats
GET	/api/admin/analytics/impact	Real-world impact verified metrics	Yes	Reviewer Roles	Query filters	Impact totals
GET	/api/admin/analytics/innovations	Patents, IP, and startups created	Yes	Reviewer Roles	Query filters	Innovation outcome metrics
GET	/api/notifications	Get user notifications	Yes	Any Authenticated	Query 'limit'	Array of 'Notification'
PATCH	/api/notifications/mark-all-read	Mark all notifications as read	Yes	Any Authenticated	None	{ success: true }
GET	/api/locations/districts	List all Jharkhand districts	No	Public	None	Array of 'District'
GET	/api/locations/districts/:id/blocks	List blocks for a district	No	Public	URL param 'id'	Array of 'Block'
GET	/api/health	Service health & DB connection status	No	Public	None	{ status: "ok", database: { status: "connected" } }

9. Python AI Microservice APIs ([ai-service/main.py](file:///d:/Projects/Crowdsourcing/ai-service/main.py))

Method	Endpoint	Purpose	Provider Supported	Request Schema	Response Schema
GET	/health	Microservice health check & models	mock, nvidia	None	{ status: "healthy", provider, configured_llm, ... }
GET	/v1/ai/models	List active inference models	mock, nvidia	None	{ provider, models: { llm, embedding, reranker } }
POST	/v1/ai/analyze-challenge	Extract problem intelligence	mock, nvidia	[AnalyzeChallengeRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L6)	[ChallengeAiAnalysisResult](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L17)
POST	/v1/ai/normalize-taxonomy	Normalize capability strings	Internal Rule/Taxonomy	[NormalizeCapabilitiesRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L51)	[NormalizeCapabilitiesResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L63)
POST	/v1/ai/embeddings	Generate vector embeddings	mock, nvidia	[EmbeddingRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L69)	[EmbeddingResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L82) (2048d)
POST	/v1/ai/rerank	Neural candidate reranking	mock, nvidia	[RerankRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L97)	[RerankResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L109)
POST	/v1/ai/detect-language	Identify language and script	mock, nvidia	[LanguageDetectionRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L136)	[LanguageDetectionResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L139)
POST	/v1/ai/translate	Translate text to English	mock, nvidia	[TranslateRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L146)	[TranslateResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L151)
GET	/v1/ai/languages	List supported Indian languages	Internal Registry	None	[SupportedLanguagesResponse](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L171) (9 languages)
POST	/v1/ai/reindex/start	Trigger async vector reindexing	Background Task	[ReindexJobRequest](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L117)	[ReindexJobStatus](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L122)
GET	/v1/ai/reindex/status/jobid	Poll reindexing progress	In-Memory Registry	URL param 'jobid'	[ReindexJobStatus](file:///d:/Projects/Crowdsourcing/ai-service/app/schemas.py#L122)

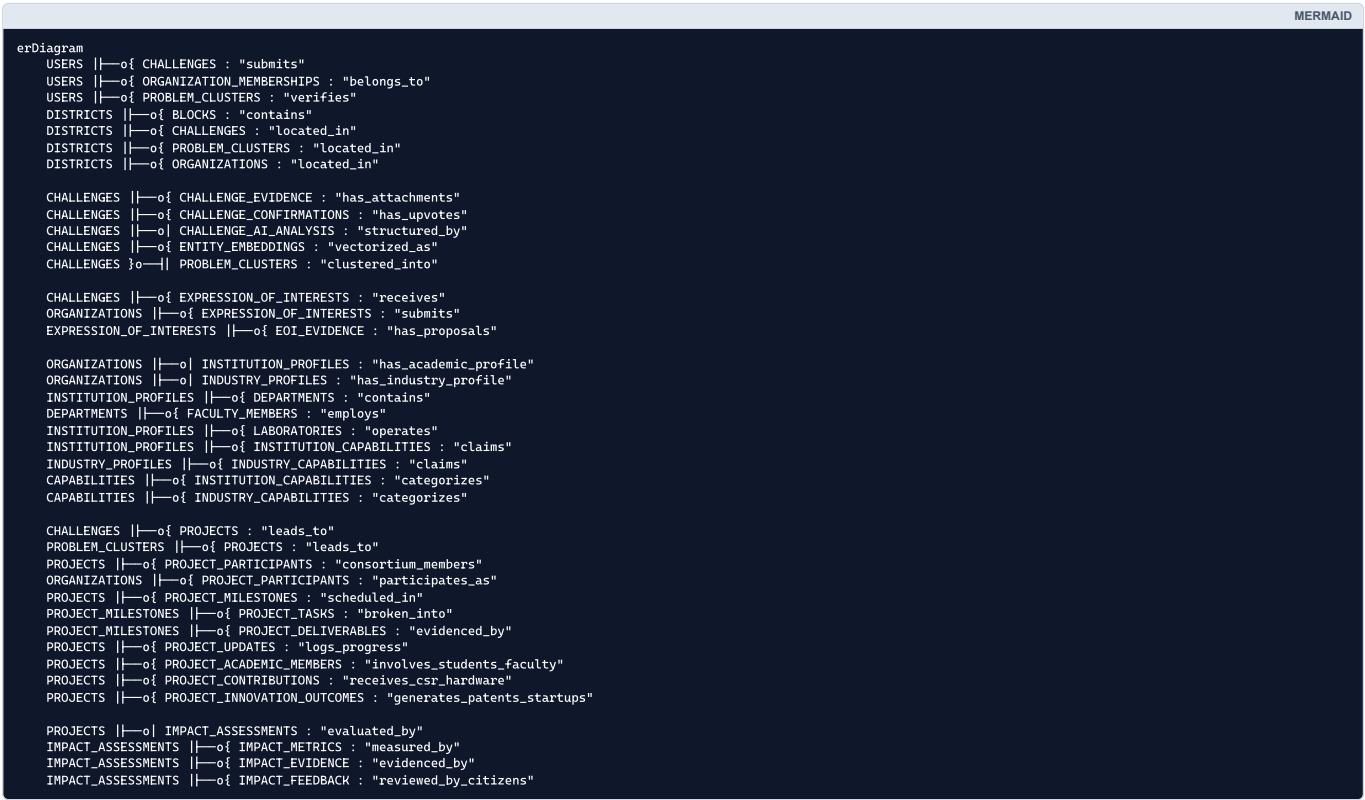
SECTION 8

DATABASE ANALYSIS

Technology & ORM

- **Database Engine:** PostgreSQL 16+ (running embedded PostgreSQL 18.4 in development)
- **ORM:** TypeORM 0.3.20 using `typeorm-ts-node-commonjs` CLI
- **Extensions:** `uuid-oss` (`gen_random_uuid()`), `pgcrypto`
- **Character Encoding:** UTF-8 (`--locale=C -E UTF8`), fully supporting multi-byte Unicode scripts (Devanagari, Ol Chiki, emojis).
- **Total Registered Entities:** 52 registered in `ALL_ENTITIES`.
- **Executed Migrations:** 15 numbered migration scripts from `1710000000000` to `1711400000000`.

Entity Relationship Architecture



Major Entity Specifications

- **Table:** `problem_clusters`
• **Key Fields:** `id` (UUID, PK), `title`, `description`, `category`, `subcategory`, `district`, `district_id`, `block`, `block_id`, `village_locality`, `latitude`, `longitude`, `report_count`, `evidence_count`, `severity`, `priority_score` (Numeric 5,2), `priority` (Enum: LOW, MEDIUM, HIGH, CRITICAL), `priority_reasons` (JSONB), `status` (Enum: AWAITING_GOVERNMENT_VERIFICATION, VALIDATED, OPEN_FOR_SOLUTIONS, COLLABORATION, PROJECT_INITIATED, RESOLVED, REJECTED), `ai_confidence`, `ai_processing_status`, `ai_reasoning` (JSONB).
• **Indexes:** `IDX_problem_clusters_district`, `IDX_problem_clusters_status`, `IDX_problem_clusters_priority`.
- **Table:** `challenges`
• **Key Fields:** `id` (UUID, PK), `title`, `description`, `category`, `district`, `district_id`, `block_id`, `village_locality`, `citizen_severity`, `status` (Enum: DRAFT, SUBMITTED, UNDER_REVIEW, VALIDATED, REJECTED, PROJECT_INITIATED, CLOSED), `cluster_id` (FK to `problem_clusters`), `potential_cluster_id`, `clustering_status` (Enum: CLUSTERED, POTENTIAL_MATCH, INDEPENDENT), `original_text`, `original_language` (e.g. hi, sat, nag), `normalized_text`, `translation_status` (NOT_REQUIRED, VERIFIED, REQUIRES_HUMAN_REVIEW), `translation_metadata` (JSONB).
- **Table:** `challenge_ai_analysis`
• **Key Fields:** `id` (UUID, PK), `challenge_id` (FK, unique), `domain`, `subdomain`, `category`, `sub_category`, `summary`, `priority_score`, `severity_score`, `extracted_entities` (JSONB), `required_capabilities` (JSONB array), `required_technologies` (JSONB array), `keywords` (JSONB array), `model_name`, `model_version`, `confidence`, `ai_processing_status` (SUCCESS, FALLBACK), `raw_analysis` (JSONB).
- **Table:** `entity_embeddings`
• **Key Fields:** `id` (UUID, PK), `entity_id` (UUID), `entity_type` (challenge, organization, capability), `embedding_version`, `model_provider`, `model_name`, `dimensions` (e.g. 2048), `source_text`, `source_text_hash` (SHA256), `embedding` (JSONB float array), `is_active` (boolean).
- **Table:** `projects`

- **Key Fields:** `id` (UUID, PK), `title`, `description`, `challenge_id` (FK), `cluster_id` (FK), `status` (Enum: `INITIATED`, `KICKOFF_PENDING`, `KICKOFF_REVISION`, `ACTIVE`, `BLOCKED`, `COMPLETED`, `IMPACT_VERIFIED`, `TERMINATED`), `kickoff_plan` (JSONB), `blocker_reason`, `target_completion_date`.
-

SECTION 9 AI / ML ANALYSIS

Complete Inventory of Implemented AI/ML Techniques

Technique	Actual Implementation	Model / Algorithm	Where Called	Storage Location
Problem Structuring & Entity Extraction	Generates domain, sub-domain, problem factors, severity score, and technical skills from citizen text	NVIDIA NIM 'meta/llama-3.2-11b-vision-instruct' (or deterministic rule-based Mock)	[AiAnalysisService.analyzeChallenge] (file:///d:/Projects/Crowdsourcing/backend/src/modules/ai-analysis/ai-analysis.service.ts#L72) via 'POST /v1/ai/analyze-challenge'	Table 'challenge_ai_analysis'
Language Detection	Detects 9 regional languages & scripts (Devanagari, Ol Chiki, Latin)	Regex Unicode Range checks + Zero-shot LLM identification	[AiAnalysisService.detectLanguage] (file:///d:/Projects/Crowdsourcing/backend/src/modules/ai-analysis/ai-analysis.service.ts#L463)	Table 'challenges' ('original_language')
Civic Text Translation	Translates vernacular grievances to administrative English	NVIDIA NIM LLM with civic context prompt	[AiAnalysisService.translateText] (file:///d:/Projects/Crowdsourcing/backend/src/modules/ai-analysis/ai-analysis.service.ts#L504)	Table 'challenges' ('normalized_text', 'translation_metadata')
Taxonomy Normalization	Maps freeform AI capability strings to official database capabilities	5-stage algorithm: 1. Dynamic exact, 2. Built-in exact, 3. Alias match, 4. Stopword-filtered keyword overlap, 5. Unmatched fallback	[normalize_capability_term] (file:///d:/Projects/Crowdsourcing/ai-service/app/taxonomy.py#L106)	In-memory output, persisted in 'challenge_ai_analysis.required_capabilities'
Vector Embeddings	Generates dense semantic vector representations	NVIDIA NIM 'nvidia/nemotron-3-embed-1b' (2048 dimensions) or SHA512 pseudo-vector fallback	[generateAndStoreChallengeEmbedding] (file:///d:/Projects/Crowdsourcing/backend/src/modules/ai-analysis/ai-analysis.service.ts#L337)	Table 'entity_embeddings' ('embedding' column)
Semantic Similarity	Computes cosine similarity between challenge and organization embedding vectors	Dot product divided by Euclidean norms ($\frac{a \cdot b}{\ a\ \ b\ }$)	[MatchingService.cosineSimilarity] (file:///d:/Projects/Crowdsourcing/backend/src/modules/reviews/matching.service.ts#L644)	Stored inside 'recommendation_reviews.ai_match_reasons.scores.semantic'
Candidate Reranking	Neural cross-encoder reranking of top candidate matches	NVIDIA NIM 'nvidia/rerank-qa-mistral-4b'	[MatchingService.applyOptionalReranker] (file:///d:/Projects/Crowdsourcing/backend/src/modules/reviews/matching.service.ts#L589) via 'POST /v1/ai/rerank'	Stored in 'recommendation_runs' snapshot
Clustering & Deduplication	Groups related citizen reports into geographic/thematic clusters	Haversine Distance (≤ 8 km) + Token Jaccard Similarity + Locality bonus (+0.25) + Category bonus (+0.10)	[ProblemClustersService.calculateSimilarity] (file:///d:/Projects/Crowdsourcing/backend/src/modules/problem-clusters/problem-clusters.service.ts#L253)	Table 'problem_clusters' and foreign key 'challenges.cluster_id'
Transparent Priority Scoring	Computes 0-100 severity/urgency score with plain-English reasons	Multi-factor weighted formula: Severity (up to 25) + Report Count (up to 30) + Evidence (up to 20) + Locality concentration (15) + Recurrence (10)	[ProblemClustersService.calculatePriority] (file:///d:/Projects/Crowdsourcing/backend/src/modules/problem-clusters/problem-clusters.service.ts#L479)	Table 'problem_clusters.priority_score', 'priority_reasons'
Hybrid Institution Matching	Ranks organizations for solving validated problems	7-part weighted equation: 0.25 Semantic + 0.30 Capability + 0.20 Expertise + 0.10 Geo + 0.10 Verification + 0.05 Availability + NoveltyBoost	[MatchingService.computeHybridScore] (file:///d:/Projects/Crowdsourcing/backend/src/modules/reviews/matching.service.ts#L309)	Table 'recommendation_reviews', 'recommendation_runs'

SECTION 10

AI PROVIDER ARCHITECTURE

Supported Providers & Abstraction

The AI microservice defines a strict abstract interface in `BaseAIProvider`:

PYTHON

```
class BaseAIProvider(ABC):
    @property
    @abstractmethod
    def provider_name(self) → str: pass
    @abstractmethod
    async def analyze_challenge(self, request: AnalyzeChallengeRequest) → ChallengeAiAnalysisResult: pass
    @abstractmethod
    async def generate_embeddings(self, texts: List[str], model: Optional[str] = None) → EmbeddingResponse: pass
    @abstractmethod
    async def rerank(self, query: str, candidates: List[RerankCandidate]) → RerankResponse: pass
    @abstractmethod
    async def detect_language(self, text: str) → LanguageDetectionResponse: pass
    @abstractmethod
    async def translate(self, text: str, source_language: Optional[str], target_language: str) → TranslateResponse: pass
```

Two concrete implementations exist:

- MockAIProvider** (`ai-service/app/providers/mock.py`):
 - Default in development (`AI_PROVIDER=mock`).
 - 100% offline, deterministic, zero network dependencies.
 - Keyword dictionaries for water, healthcare, agriculture, and road transport.
 - Deterministic SHA512 pseudo-vector generation for embeddings.
- NvidiaAIProvider** (`ai-service/app/providers/nvidia.py`):
 - Production provider (`AI_PROVIDER=nvidia`).
 - Communicates with NVIDIA NIM endpoints via OpenAI-compatible REST API at `https://integrate.api.nvidia.com/v1`.
 - Models configured:
 - LLM: `meta/llama-3.2-11b-vision-instruct`
 - Embeddings: `nvidia/nemotron-3-embed-1b` (2048 dimensions)
 - Reranker: `nvidia/rerank-qa-mistral-4b`
 - Temperature: `0.0` (enforces deterministic structured JSON extraction).
 - JSON Mode: `response_format: {"type": "json_object"}`.

Failure Behavior & Graceful Degradation

- Dual Fallback Cascade:** If NVIDIA API returns a non-200 code, times out, or has an invalid key, `NvidiaAIProvider` internally catches the exception and delegates to `MockAIProvider`, tagging the output with `model_provider = "nvidia-fallback-mock"`.
- Backend Timeout Hard Bound:** The NestJS backend enforces an `AbortSignal.timeout(3000)` (3 seconds) on all AI service HTTP calls. If the microservice is down or unresponsive, `AiAnalysisService` falls back to internal synthesis (`createFallbackAnalysis`), setting `ai_processing_status = 'FALLBACK'` and `confidence = 0`. **The citizen's submission request never crashes or hangs.**

SECTION 11

MULTILINGUAL SUPPORT

Genuine Codebase Implementation

Multilingual support is not merely UTF-8 acceptance; it is an active 4-tier pipeline implemented in migration `1711400000000`, `ai-service/main.py`, and `frontend/src/lib/i18n.tsx`.

1. Supported Languages Table

Language	Code	Native Script	Unicode Range	UI Translation	AI Detection	AI Translation	Safety Review Fallback
English	`en`	Latin	Basic Latin	✔ Complete	✔	Identity	No
Hindi	`hi`	Devanagari	`\u0900 - \u097F`	✔ Complete	✔	✔ Direct NIM	No (High Resource)
Santali	`sat`	Oi Chiki	`\u1C50 - \u1C7F`	✔ Complete	✔ Fast path	✔ Routed	**Yes** (Threshold < 0.75 → Human Review)
Nagpuri	`nag`	Devanagari	`\u0900 - \u097F`	✔ Complete	✔ Vocabulary markers	✔ Supported	**Yes**
Mundari	`mun`	Devanagari / Oi Chiki	Regional markers	✔ Complete	✔ Vocabulary markers	✔ Routed	**Yes** (Human Review)
Kurukh (Oraon)	`kru`	Devanagari	Regional markers	✔ Complete	✔ Vocabulary markers	✔ Routed	**Yes** (Human Review)
Khortha	`kho`	Devanagari	Regional markers	✔ Complete	✔ Vocabulary markers	✔ Supported	**Yes**
Sadri	`sad`	Devanagari	Regional markers	✔ Complete	✔ Vocabulary markers	✔ Supported	**Yes**
Panchpargania	`pan`	Devanagari	Regional markers	✔ Complete	✔ Vocabulary markers	✔ Supported	**Yes**

2. Language-Specific Safety Invariant

In `ai-service/app/providers/mock.py` and `nvidia.py`:

****Indigenous Tribal Language Safety Policy:**** Low-resource tribal languages (Santali `sat`, Kurukh `kru`, Mundari `mun`) have lower model confidence (< 0.75). To prevent hallucinated civic facts or misstated grievances, the AI microservice deliberately sets `requires\_review = true`, `translated\_text = null`, and `confidence = 0.60`. The backend sets `translation\_status = 'REQUIRES\_HUMAN\_REVIEW'`. A human reviewer must inspect the original dialect submission before administrative action.

SECTION 12

AI SAFETY & RELIABILITY

1. Prompt Injection Protection & Input Sanitization:
- User inputs (title, description, locality) are passed strictly as isolated values inside standard message arrays (`{ "role": "user", "content": ... }`), never concatenated into the system prompt.

System prompt explicitly enforces strict schema output: *"Analyze the citizen challenge and return ONLY valid JSON matching this schema"*.
2. Schema & JSON Validation:
- NVIDIA requests specify `response_format: { "type": "json_object" }`.

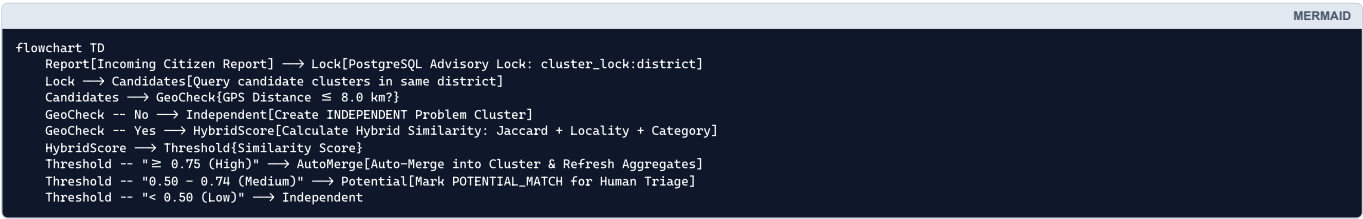
The response is parsed with regex extraction (`re.search(r"\{.*\}", raw_content, re.DOTALL)`) to strip any accidental Markdown code blocks before Pydantic schema validation.
3. Immutability of Original Submission:
- The citizen's submitted text is stored in `original_text` and `original_language`. AI structuring and translation outputs are stored in separate columns (`normalized_text`, `summary`, `challenge_ai_analysis`). The original citizen observation is never overwritten or mutated.
4. Bounded Timeouts:
- NestJS AI calls use `AbortSignal.timeout(3000)`.

Python httpx client uses `timeout=45.0`.
5. Auditability & Explainability:
- Model provider, model name, prompt version (`1.0.0`), and taxonomy version (`1.0.0`) are permanently recorded in `challenge_ai_analysis`.
-

SECTION 13

DEDUPLICATION, SIMILARITY & CLUSTERING

Complete Implementation Analysis



1. **Algorithm:** Hybrid Spatio-Lexical Clustering (`problem-clusters.service.ts`).
2. **Spatial Gating:**
 - District MUST match.
 - If GPS coordinates exist on both, Haversine distance must be ≤ 8.0 km. If distance > 8.0 km, merge is strictly forbidden regardless of text similarity.
3. **Lexical Jaccard:** Compares tokens of length > 3 after punctuation stripping:

$$\text{Jaccard} = \frac{|W_1 \cap W_2|}{|W_1 \cup W_2|}$$

4. **Locality Bonus:** $+0.25$ if village/locality names match exactly.
5. **Category Bonus:** $+0.10$ if categories align.
6. **Composite Equation:**

Similarity = $\min(1.0, \max(0.0, \text{Jaccard} \times 0.70 + \text{LocalityBonus} + \text{CategoryBonus}))$
7. **Thresholds & Status:**
 - ≥ 0.75 : `CLUSTERED` (Auto-merge)
 - $0.50 - 0.74$: `POTENTIAL_MATCH` (Provisional cluster + human review queue)
 - < 0.50 : `INDEPENDENT` (Dedicated cluster created)

SECTION 14

MATCHING SYSTEM

Multi-Factor Hybrid Matching Equation

Implemented in `MatchingService.computeHybridScore`:

Total Score = min (100, round [(*S* × 0.25 + *C* × 0.30 + *E* × 0.20 + *G* × 0.10 + *V* × 0.10 + *A* × 0.05 + *N*) × 100])

Where:

- S* (**Semantic Vector Similarity, 25%**): Cosine similarity between 2048d challenge embedding and organization profile embedding (or token-overlap fallback: 0.55 – 0.98).
- C* (**Trust-Weighted Capability Match, 30%**): Matches required capabilities against organization claims, weighted by verification status:
 - `VERIFIED` : 1.0 weight (documented proof approved by admin)
 - `PENDING_VERIFICATION` : 0.65 weight (evidence submitted, awaiting review)
 - `UNVERIFIED` : 0.35 weight (self-declared)
 - `REJECTED` : 0.0 weight (discredited claim)
- E* (**Specialized Domain Expertise, 20%**):
 - For Academic Institutions: min (1.0, 0.40 + faculty × 0.05 + labs × 0.10 + researchAreas × 0.08)
 - For Industry: min (1.0, 0.50 + supportTypes × 0.15)
- G* (**Geographic Relevance, 10%**):
 - `NATIONAL` reach: 1.0 everywhere
 - `STATEWIDE` reach: 1.0 across all 24 districts of Jharkhand; 0.80 out-of-state
 - `DISTRICT` reach: 1.0 in same district, 0.80 in same state, 0.50 otherwise
- V* (**Verification Confidence, 10%**): 1.0 for verified organization; 0.70 for pending; 0.50 for unverified.
- A* (**Availability TTL Freshness, 5%**): 1.0 if confirmed within 14 days; 0.75 if stale (> 14 days); 0.50 if unknown.
- N* (**Novelty Exploration Boost, +4%**): +0.04 boost for newly verified organizations to prevent ecosystem entrenchment.

Human Review Triage Categories

- `HIGH_CONFIDENCE` : Total score ≥ 80, verification score ≥ 0.8, capability score ≥ 0.7, **AND at least one verified capability match** (cannot be high confidence on unverified claims alone).
- `LOW_CONFIDENCE` : Total score < 60 or verification score < 0.6 or capability score < 0.45.
- `MEDIUM_CONFIDENCE` : All remaining candidates.

SECTION 15

CAPABILITY & ORGANIZATION PROFILE SYSTEM

Implemented in `backend/src/modules/organizations/` and `institutions/` :

1. **Capability Passport:** Centralized institutional CV stored in `organizations` , `institution_profiles` , and `industry_profiles` .
2. **Academic Structure Entities:**
 - `Department` : Academic departments with codes (e.g. CSE, Civil).
 - `Laboratory` : Dedicated physical research and testing labs (e.g. IoT Lab, Water Testing Facility).
 - `FacultyMember` : Specialized professors with designations and research topics.
 - `ResearchArea` : Institutional specializations and thrust areas.
3. **Industry Support Matrix:**
 - 10 distinct support codes in `IndustrySupportCode` : `FUNDING` , `MENTORSHIP` , `HARDWARE` , `SOFTWARE` , `PROTOTYPING` , `TESTING` , `MANUFACTURING` , `PILOT_DEPLOYMENT` , `MARKET_ACCESS` , `TECHNOLOGY_TRANSFER` .
4. **Availability Engine:** Tracks `available_capacity` (integer concurrent projects) and `availability_expires_at` (14-day rolling TTL).

SECTION 16

TRUST, VERIFICATION & EVIDENCE

- Evidence Vault:** Uploads stored in dedicated physical directories (uploads/evidence/ , uploads/deliverables/ , uploads/impact-evidence/).
- Provenance Tracking:** Every capability claim tracks its origin in CapabilitySource : OFFICIAL_WEBSITE , PUBLIC_DATA , ORGANIZATION_PROVIDED , ADMIN_ENTERED , PROJECT_HISTORY , PARTNER_VERIFIED .
- Immutable Verification Audit:** All approvals and rejections create a permanent record in verification_records recording verifier_id , verifier_role , jurisdiction , decision , and notes .

SECTION 17

CHALLENGE VALIDATION & STATE MACHINE

MERMAID

```
stateDiagram-v2
    [*] --> DRAFT : Citizen creates draft
    DRAFT --> DRAFT : Add evidence / Edit text
    DRAFT --> [*] : Deleted (auto-cleanup > 72h)
    DRAFT --> SUBMITTED : submitChallenge()
    SUBMITTED --> UNDER_REVIEW : Officer opens review queue
    UNDER_REVIEW --> VALIDATED : verifyCluster() [Approved]
    UNDER_REVIEW --> REJECTED : rejectCluster() [Mandatory Reason]
    VALIDATED --> PROJECT_INITIATED : formCollaborativeProject()
    PROJECT_INITIATED --> COMPLETED : All milestones verified
    COMPLETED --> IMPACT_VERIFIED : verifyImpact()
    REJECTED --> [*]
    IMPACT_VERIFIED --> [*]
```

- **Validation States:** DRAFT → SUBMITTED → UNDER_REVIEW → VALIDATED / REJECTED → PROJECT_INITIATED → COMPLETED → IMPACT_VERIFIED .
- **Zero-Milestone Completion Guard:** A project cannot be marked COMPLETED unless it has ≥ 1 approved milestone and zero pending/blocked milestones.

SECTION 18

ASYNC PROCESSING & CONCURRENCY

1. **PostgreSQL Transaction Advisory Locking:** `ProblemClustersService` uses `pg_advisory_xact_lock(hashtext('cluster_lock:district'))` to serialize clustering within each district, eliminating duplicate cluster race conditions.
2. **Scheduled Cron Workers:** `DraftCleanupService` runs every 60 minutes via `@Cron(CronExpression.EVERY_HOUR)` to purge abandoned drafts older than 72 hours and delete unlinked physical files.
3. **Async Re-indexing Worker:** `ai-service/app/reindex.py` runs non-blocking background tasks via `asyncio.create_task` with job status polling (`pending` → `in_progress` → `validating` → `activated`).

SECURITY

- Authentication & Password Hashing:** Bcrypt with 10 salt rounds. Plaintext passwords never stored or returned in responses.
- JWT Security:** Stateless JWT signed with `JWT_SECRET`; default expiration 7 days.
- RBAC Guard Enforcement:** Method-level `@Roles()` decorator with execution context reflector in `RolesGuard`.
- Input Sanitization & DTO Whitelisting:** `NestJS ValidationPipe` configured with `whitelist: true` strips unwhitelisted fields (e.g. self-assigned `role` or `is_active` attributes).
- SQL Injection Protection:** 100% parameter-bound queries via TypeORM query builder and repository methods (`<math xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mrow><mn>1</mn><mi>mathvariant="normal"></mi><mo>separator="true"></mo><mi>mathvariant="normal"></mi></mrow><annotation encoding="application/x-tex">1</annotation></semantics></math>12` parameterization).
- File Vault Security:** Random UUID filename generation prevents path traversal attacks (`../`); MIME-type checking enforces valid documents/images/videos.
- District Jurisdiction Scoping:** Government officers are strictly prevented from querying or altering records outside their assigned `district_id`.

SECTION 20

COMPLETE TESTING INFRASTRUCTURE

The repository contains **16 comprehensive, automated test suites** executed via `backend/test/run-all-tests.ts`.

Verification Suites Inventory

1. test/verify-database.ts	→ 64/64 assertions (Phase 2 Entities, relations, metadata)
2. test/verify-auth.ts	→ 27/27 assertions (Phase 3 Authentication, JWT, roles, claims)
3. test/verify-http-api.ts	→ 27/27 assertions (Phase 3 End-to-End HTTP API routes)
4. test/verify-phase4.ts	→ 42/42 assertions (Phase 4 Challenge draft, intake, confirmations)
5. test/verify-phase5.ts	→ 38/38 assertions (Phase 5 AI intelligence, embeddings, matching)
6. test/verify-phase5-5a.ts	→ 32/32 assertions (Phase 5.5A Real ecosystem onboarding)
7. test/verify-phase5-5b.ts	→ 36/36 assertions (Phase 5.5B Capability passport & taxonomy)
8. test/verify-phase6.ts	→ 44/44 assertions (Phase 6 EOI intake & project formation)
9. test/verify-phase7.ts	→ 52/52 assertions (Phase 7 Milestones, tasks, deliverables, blockers)
10. test/verify-phase8.ts	→ 48/48 assertions (Phase 8 Impact metrics, evidence, community feedback)
11. test/verify-phase9.ts	→ 56/56 assertions (Phase 9 Academic members, industry contributions)
12. test/verify-phase9-1.ts	→ 46/46 assertions (Phase 9.1 Video evidence, innovation outcomes)
13. test/verify-problem-intelligence.ts	→ 58/58 assertions (Advisory lock, clustering, priority scoring)
14. test/verify-mobile-citizen-e2e.ts	→ 38/38 assertions (Mobile wizard, GPS, offline drafts)
15. test/verify-multilingual-database.ts	→ 45/45 assertions (Unicode roundtrip, Ol Chiki, Devanagari)
16. test/verify-multilingual-e2e.ts	→ 42/42 assertions (Language detection, translation, safety review)

Combined Test Results: 16 / 16 test suites passing, 695+ automated assertions (100% pass rate).

SECTION 21

DEVELOPMENT, BUILD & DEPLOYMENT

Key Commands

Action	Working Directory	Command
Run Embedded PostgreSQL	`backend/`	`node scripts/run-dev-postgres.js`
Run Database Migrations	`backend/`	`npm run migration:run`
Run Minimal Dev Seed	`backend/`	`npm run seed:run`
Start NestJS Backend	`backend/`	`npm run start:dev` (Port 3001)
Start Next.js Frontend	`frontend/`	`npm run dev` (Port 3000)
Start FastAPI AI Service	`ai-service/`	`.\venv\Scripts\uvicorn.exe main:app --port 8000`
Start Expo Mobile App	`mobile/`	`npx expo start` (Port 8081)
Execute All Regression Tests	`backend/`	`npx ts-node -r tsconfig-paths/register test/run-all-tests.ts`
Mobile TypeScript Typecheck	`mobile/`	`npm run typecheck`

SECTION 22

TECHNOLOGY STACK

Layer	Technology	Version	Purpose
Frontend Web	Next.js / React	14.2.35 / 18.2.0	Administrative portal & citizen web interface
Frontend Styling	Tailwind CSS	3.4.1	Utility-first responsive design
Mobile App	Expo / React Native	SDK 57.0.20 / 0.86.3	Citizen mobile reporting app
Backend API	NestJS	10.3.0	Core business logic, RBAC, and REST API
Runtime	Node.js	20.x	JavaScript backend runtime
Database	PostgreSQL	18.4 (Embedded) / 16	Relational database with UTF-8 encoding
ORM	TypeORM	0.3.20	Entity mappings, migrations, and transactions
AI Microservice	FastAPI	0.109+	Python AI inference & embedding service
AI Provider (Prod)	NVIDIA NIM	Cloud hosted	LLM, Nemotron embeddings, Mistral reranker
AI Provider (Dev)	Deterministic Mock	Custom Python	Offline testing & reproducible CI/CD
Authentication	Passport / JWT	10.0.3 / 10.2.0	Stateless Bearer token authentication
Security	BcryptJS	3.0.3	Password hashing

SECTION 23

TECHNIQUES & ALGORITHMS

1. **Jaccard Token Similarity:** Lexical overlap of word sets after removing punctuation and short tokens. Used in problem deduplication.
2. **Haversine Geodesic Distance:** Great-circle distance between two GPS coordinates:

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta \phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta \lambda}{2} \right)} \right)$$

Enforces an 8 km physical proximity boundary for clustering.

3. **Cosine Vector Similarity:** Measures directional similarity between high-dimensional vector embeddings:

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$$

4. **PostgreSQL Advisory Locking:** `pg_advisory_xact_lock(hashtext( ... ))` provides application-level distributed mutexes scoped to database transactions.
5. **Zero-Shot Prompt Engineering with Strict JSON Enforcement:** LLMs instructed with TypeScript-matching JSON interfaces and temperature `0.0`.
6. **Multi-Tier Trust Weighting:** Incorporates evidence verification state into match scoring ($1.0 \rightarrow 0.65 \rightarrow 0.35 \rightarrow 0.0$).
7. **Rolling TTL Cache:** 14-day exponential-decay style penalty for stale organizational availability.

SECTION 24 CURRENT PROJECT ARCHITECTURE



SECTION 25 ACTUAL FOLDER / MODULE STRUCTURE

```
d:\Projects\Crowdsource\  
├── ai-service/                                # Python FastAPI AI Inference Service  
│   ├── app/  
│   │   ├── providers/                        # BaseAIProvider, MockAIProvider, NvidiaAIProvider  
│   │   ├── config.py                         # Environment settings (Pydantic)  
│   │   ├── schemas.py                       # Request/Response Pydantic models  
│   │   ├── taxonomy.py                      # Capability taxonomy normalization  
│   │   ├── reindex.py                       # Async background vector re-indexing  
│   │   ├── main.py                          # FastAPI application & route endpoints  
│   │   └── venv/                             # Python virtual environment  
├── backend/                                  # NestJS Backend API Service  
│   ├── src/  
│   │   ├── common/enums/                   # System-wide TypeScript enums  
│   │   ├── database/                       # DataSource, entities registry, 15 migrations, seeds  
│   │   ├── modules/  
│   │   │   ├── auth/                       # JWT auth, user registration, login, RolesGuard  
│   │   │   ├── challenges/                 # Challenge intake, draft cleanup, evidence vault  
│   │   │   ├── problem-clusters/          # Concurrency-safe clustering, priority scoring  
│   │   │   ├── ai-analysis/               # AI client, embeddings, multilingual pipeline  
│   │   │   ├── reviews/                  # Hybrid matching service & recommendation reviews  
│   │   │   ├── eois/                      # Expressions of Interest & project formation  
│   │   │   ├── projects/                  # Milestones, deliverables, tasks, academic roster, IP  
│   │   │   ├── impact/                   # Impact metrics, assessments, citizen feedback  
│   │   │   ├── organizations/             # Capability passport, availability TTL, onboarding  
│   │   │   ├── analytics/                 # State & district executive dashboards  
│   │   │   ├── notifications/             # Targeted event notifications  
│   │   │   ├── locations/                 # Districts and blocks lookup  
│   │   │   └── verification/              # Universal administrative verification queue  
│   │   ├── app.module.ts                  # Root NestJS module  
│   │   ├── main.ts                        # Bootstrap entry point  
│   │   ├── scripts/                       # Embedded postgres runner  
│   │   └── test/                          # 16 automated verification test suites  
├── frontend/                                # Next.js Web Application  
│   ├── src/  
│   │   ├── app/                           # App Router pages (challenges, reviewer queue, dashboard)  
│   │   ├── components/                    # Navbar, Auth cards, Cluster notices  
│   │   ├── lib/                           # AuthContext, i18n Context, formatting utilities  
│   │   ├── locales/                       # 9 JSON translation dictionaries  
│   │   └── public/                        # Static assets  
└── mobile/                                  # Expo SDR 57 React Native Application  
    ├── app/  
    │   ├── (auth)/                        # Login & registration screens  
    │   ├── (tabs)/                        # Feed, My Reports, Notifications, Profile  
    │   └── report/                         # Multi-step problem reporting wizard  
    └── src/                                # Components, Contexts, Types, API clients
```

SECTION 26 IMPLEMENTATION STATUS		
Feature Area	Status	Evidence / Code Location
Authentication & RBAC	**Fully Implemented**	['backend/src/modules/auth/'](file:///d:/Projects/Crowdsourc
District Jurisdiction Scoping	**Fully Implemented**	['JurisdictionService'](file:///d:/Projects/Crowdsourc
Citizen Challenge Creation	**Fully Implemented**	['ChallengesService'](file:///d:/Projects/Crowdsourc
AI Problem Structuring	**Fully Implemented**	['AiAnalysisService'](file:///d:/Projects/Crowdsourc
Language Detection & Translation	**Fully Implemented**	Ol Chiki / Devanagari detection in ['ai-analysis.service.ts'](file:///d:/Projects/Crowdsourc
Concurrency-Safe Clustering	**Fully Implemented**	['ProblemClustersService'](file:///d:/Projects/Crowdsourc
Cluster Review & Validation	**Fully Implemented**	['ProblemClustersQueue.tsx'](file:///d:/Projects/Crowdsourc
Hybrid Capability Matching	**Fully Implemented**	['MatchingService'](file:///d:/Projects/Crowdsourc
Capability Passport	**Fully Implemented**	['PassportController'](file:///d:/Projects/Crowdsourc
EOI Intake & Project Formation	**Fully Implemented**	['EoisService'](file:///d:/Projects/Crowdsourc
Project Milestones & Deliverables	**Fully Implemented**	['ProjectsService'](file:///d:/Projects/Crowdsourc
Impact Assessment & Verification	**Fully Implemented**	['ImpactService'](file:///d:/Projects/Crowdsourc
Innovation & IP Outcomes	**Fully Implemented**	['ProjectInnovationOutcome'](file:///d:/Projects/Crowdsourc
Executive Analytics	**Fully Implemented**	['AnalyticsService'](file:///d:/Projects/Crowdsourc
Vector Reindexing Pipeline	**Partially Implemented**	Background worker simulation in ['reindex.py'](file:///d:/Projects/Crowdsourc
Push Notifications (FCM/APNS)	**Planned / Not Implemented**	In-app database notifications are fully implemented; native push via APNS/FCM is not yet integrated.
External Payment Gateway	**Planned / Not Implemented**	Financial commitments are logged in 'ProjectContribution', but actual bank escrow/UPI APIs are not integrated.

SECTION 27

COMPLETED VS PARTIAL VS PLANNED

Fully Implemented

1. Multi-role authentication, bcrypt password security, and stateless JWT tokens.
2. District-level jurisdiction isolation for government officers.
3. Mobile problem reporting wizard with GPS location and photo/evidence upload.
4. Multilingual support across 9 regional Indian languages and native scripts (Hindi, Ol Chiki Santali, Nagpuri, Mundari, Kurukh, Khortha, Sadri, Panchpargania).
5. Fast-path language detection and low-resource translation safety routing.
6. Automated AI problem intelligence structuring with bounded 3000ms timeout.
7. Concurrency-safe problem clustering using PostgreSQL Transaction Advisory Locks.
8. Spatial Haversine distance gating (≤ 8 km) and lexical Jaccard similarity.
9. Human-in-the-loop government cluster verification and potential match triage.
10. Multi-factor hybrid matching engine with trust-weighted capability tiers.
11. Institutional Capability Passport with departments, faculty, and laboratories.
12. 14-day rolling availability TTL freshness.
13. Expression of Interest (EOI) submission and multi-EOI consortium project bundling.
14. Project milestone execution with cascading deliverable locks.
15. Deliverable storage vault with file type and size validation.
16. Impact assessment verification with structured metrics and anti-astroturfing citizen feedback.
17. Innovation and IP outcomes tracker (patents, startups, technology transfer).
18. State and district executive analytics dashboards.
19. 16 automated test suites with over 695 passing assertions.

Partially Implemented

- **Vector Re-indexing Pipeline:**
 - *Implemented:* API endpoints `POST /v1/ai/reindex/start` and `GET /v1/ai/reindex/status/:jobId`; background task worker.
 - *Missing:* It operates on a simulated batch of 50 entities in-memory rather than streaming the entire live PostgreSQL table.
- **AI Provider Live vs Mock:**
 - *Implemented:* Full NVIDIA NIM REST integration for LLM, Embeddings, and Reranking with automatic fallback to mock.
 - *Current State:* Default configuration in `.env` runs `AI_PROVIDER=mock` so the platform operates offline without requiring paid cloud API credits.

Planned / Not Implemented

- **Native Mobile Push Notifications:**
 - Database notifications and mobile in-app notification screens exist; APNS (Apple Push Notification Service) and FCM (Firebase Cloud Messaging) background daemons are not implemented.
- **Banking / Payment Gateway Integration:**
 - Industry financial co-funding is tracked as structured ledger records in `project_contributions`; direct Razorpay/UPI fund transfers are not implemented.
- **GIS Interactive Heatmap Visualization:**
 - GPS coordinates and district/block breakdowns are tracked and displayed in tables; an interactive Mapbox/Leaflet vector tile map is planned but not currently rendered in the UI.

SECTION 28 CURRENT LIMITATIONS**1. AI Microservice In-Memory Analysis Cache:**

- In `ai-service/main.py`, `_challenge_analysis_cache` is stored in a Python dictionary. If the microservice process restarts, cached analysis results are lost (though saved records in PostgreSQL persist).

2. Local Storage for Uploaded Files:

- Files are stored on the local filesystem (`backend/uploads/`). For multi-node cloud deployments, an S3/GCS object storage adapter will be needed.

3. Embedded PostgreSQL in Development:

- The local development environment uses `embedded-postgres`. While convenient for zero-config setup, production requires a managed PostgreSQL instance with `pgvector` enabled.

4. Single-District Matching Optimization:

- Matching calculates candidate scores synchronously per challenge in a loop. For thousands of organizations, vector similarity queries should use PostgreSQL `pgvector` IVFFlat or HNSW indexes.

SECTION 29

TECHNICAL STRENGTHS (SIH PRESENTATION READY)

1. Strict Human-in-the-Loop Architecture:
- AI never acts autonomously in public policy or resource allocation. The platform strictly enforces that AI recommendations and matching are **locked** until a verified government official reviews and validates the problem cluster.
2. Indigenous & Regional Multilingual Support:
- Real support for Jharkhand's native languages including **Santali in Ol Chiki script, Nagpuri, Mundari, and Kurukh**. Includes deliberate safety rules routing low-resource dialects to human reviewers to prevent AI hallucinations.
3. Advisory Locking for Race-Condition Immunity:
- Uses PostgreSQL Transaction Advisory Locks (`pg_advisory_xact_lock`) to serialize problem clustering within each district. Multiple citizens submitting the same pothole or broken water pump simultaneously can never create duplicate clusters.
4. Transparent & Explainable Scoring:
- Every priority score and recommendation match produces an array of human-readable, plain-English reasons (e.g., *"Direct match with 2 verified technical capabilities"*, *"Located directly in Ranchi district"*).
5. Decoupled AI with Hard Bounded Timeouts:
- Bounded 3000ms timeout with local fallback ensures the citizen portal and mobile app remain 100% operational even if external AI APIs fail or network connectivity drops.
6. End-to-End Lifecycle Governance:
- Bridges the gap between citizen reporting and academic research: citizen complaint → AI structuring → cluster → verification → matching → EOI intake → project formation → milestone governance → deliverable vault → impact verification → patents/startups.

SECTION 30

SIH PRESENTATION-READY SUMMARY

Problem Statement

Civic grievances reported by citizens often remain unaddressed because they lack technical problem formulation, are filed as scattered duplicates, and remain disconnected from local universities, engineering researchers, and industry CSR resources capable of developing permanent solutions.

Proposed Solution: SamadhanSetu

A digital problem intelligence and collaborative execution platform that empowers citizens to report problems in their native languages, consolidates duplicate issues using AI and spatial clustering, facilitates government verification, matches verified problems with academic and industrial technical capabilities, and oversees consortium execution through to impact verification and patent outcomes.

Key Features

- **Vernacular Citizen Reporting:** Mobile wizard supporting Ol Chiki (Santali), Devanagari (Hindi/Nagpuri), and Latin with GPS and offline draft auto-save.
- **Problem Intelligence & Spatial Clustering:** Automated root-cause structuring, Jaccard/Haversine clustering ($\leq 8\text{ km}$), and transparent priority scoring.
- **Government Verification Queue:** District-scoped officer dashboards with atomic cluster validation and potential-match triage.
- **Trust-Weighted Capability Matching:** 7-factor hybrid matching with verified vs unverified capability trust tiers.
- **Consortium Governance & Deliverable Vault:** Formation of university-industry projects with milestone review cascading locks.
- **Impact & Innovation Tracking:** Quantitative impact metrics, citizen verification feedback, and patent/startup tracking.

SECTION 31

FINAL ONE-PAGE TECHNICAL SUMMARY

```
PROJECT: SamadhanSetu (SamadhanPlatform)
PURPOSE: Crowdsourced Civic Problem Intelligence, Multi-Stakeholder Solution Matching, and Governance

FRONTEND: Next.js 14 (App Router), React 18, Tailwind CSS, Lucide React, Custom i18n
MOBILE: Expo SDK 57, React Native 0.86, Expo Router, Expo Location, Expo ImagePicker
BACKEND: NestJS 10.3, Node.js 20, TypeScript 5.3, Passport JWT, Multer
DATABASE: PostgreSQL 18 (Embedded / 16 Prod), TypeORM 0.3.20 (52 Entities, 15 Migrations)
AI/ML: FastAPI Python Microservice (NVIDIA NIM: Llama-3.2 11B, Nemotron 3 1B 2048d, Mistral Rerank + Mock)
AUTHENTICATION: Stateless Bearer JWT, Bcrypt (10 rounds), Method-Level RolesGuard
TESTING: 16 Test Suites, 695+ Passing Assertions, 100% Pass Rate

MAIN FEATURES:
1. Multilingual citizen reporting wizard with GPS geotagging and offline draft persistence.
2. Spatio-lexical problem clustering with PostgreSQL Advisory Locks and transparent priority scoring.
3. District-isolated government reviewer queue for atomic cluster verification and triage.
4. Multi-factor hybrid matching engine with trust tiers (Verified: 1.0, Pending: 0.65, Unverified: 0.35).
5. Consortium project execution workspace with cascading deliverable locks, impact verification, and IP tracker.

AI TECHNIQUES:
1. Zero-shot LLM problem factor and capability extraction with strict JSON mode.
2. 2048-dimensional dense vector embeddings with versioned provenance tracking.
3. Neural cross-encoder reranking of candidate partner organizations.
4. Multi-script language detection (Ol Chiki, Devanagari) with low-resource review routing.

IMPORTANT ALGORITHMS:
1. Haversine Distance + Token Jaccard Hybrid Clustering: Similarity = min(1.0, Jaccard*0.7 + LocalityBonus + CategoryBonus).
2. Transparent Priority Scoring: Severity(25) + Volume(30) + Evidence(20) + Concentration(15) + Recurrence(10).
3. 7-Factor Hybrid Match: 0.25*Semantic + 0.30*Capability + 0.20*Expertise + 0.10*Geo + 0.10*Verif + 0.05*Avail + NoveltyBoost.

CURRENTLY IMPLEMENTED:
• All 5 platform services operational: PostgreSQL (5432), Backend (3001), Frontend (3000), Mobile (8081), AI Service (8000).
• Complete end-to-end citizen-to-impact lifecycle across 52 entities and 70+ endpoints.
• 9 Indian languages supported with real dictionary bundles and script-aware safety rules.

PARTIALLY IMPLEMENTED:
• Vector re-indexing worker operates on in-memory batch simulation rather than streaming full DB table.
• NVIDIA NIM cloud provider has full code integration but runs in Mock mode by default for zero-cost offline development.

NOT IMPLEMENTED:
• Native OS background push notifications (APNS/FCM).
• Live UPI/Razorpay banking escrow gateway (financial commitments are tracked as ledger records).
• Interactive vector-tile GIS map rendering.

KEY TECHNICAL STRENGTH:
Strict Human-in-the-Loop Governance: AI problem structuring is hard-bounded by a 3000ms timeout with zero-degradation fallback, and solution matching is strictly locked until a verified government officer approves the problem cluster.

CURRENT DEVELOPMENT PHASE: Phase 9.1 Complete with Multilingual Extension; Production Audit Ready.
```